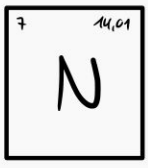
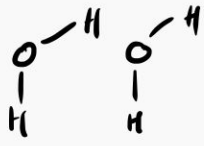
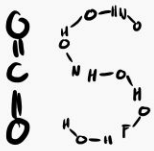
Chemistry by

DJ

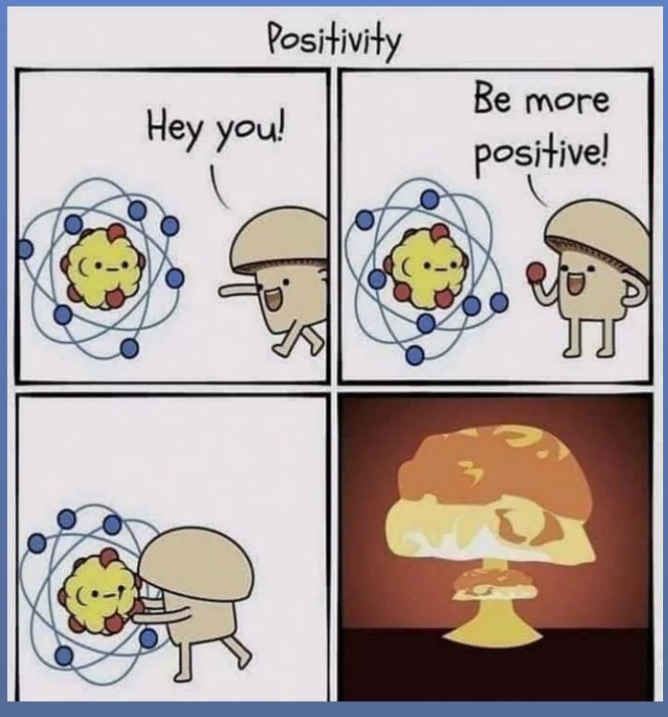








Quentin Türles Chemistry Summary

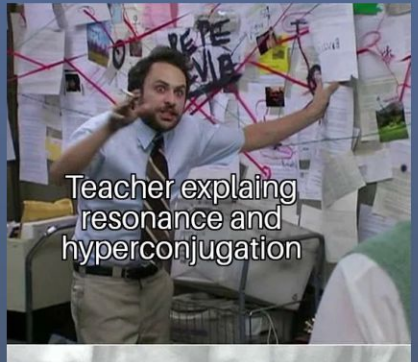
















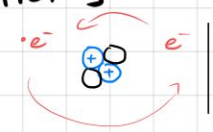




Mixtures

if uniform — homogeneous
if nonuniform — heterogeneous

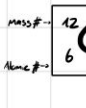
Atoms



$m_{e^-} = 9{,}11 \cdot 10^{-31}\,kg$
$m_n = 1{,}67 \cdot 10^{-27}\,kg$
$m_p = 1{,}67 \cdot 10^{-27}\,kg$

chemical properties of Atom set by $\#e^-$

Isotopes (same element with different # neutrons)

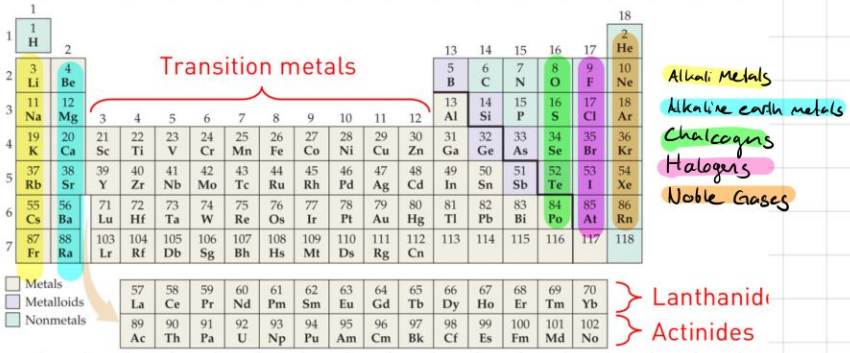


Atomic weight 12u | $u \approx 1{,}66053 \cdot 10^{-27}\,kg$

↓ set by occurrences

C: 0,98 · 12u + 0,017 · 13,0035u + trace
(^{12}C) (^{13}C) (^{14}C)

PSE



Molecular weight (Mw)

$$M_W = \frac{m}{n} \Leftrightarrow n = \frac{m}{M_W}$$

$\frac{g}{mol} = u$

Book p. 117 - 123

Naming of Ionic Compounds

Cations

pos. charge — Fe^{2+}, Fe^{3+} iron(II) ion / iron(III) ion

nonmetal molecules Cations: -ium
NH_4^+ ammonium

Anions (-)

- monoatomic/polyatomic -ide
  O^{2-} oxide OH^- Hydroxide
- polyatomic with O -ate/-ite : oxanions
  (-ate: normaler Status; -ite: one O fewer than "ate")

NO_3^- Nitrate
NO_2^- Nitrite
ClO_4^- perchlorate
ClO_3^- chlorate
ClO_2^- chlorite
ClO^- hypochlorite

-per / hypo: one more O than ate; even less O than "ite"

per... ate > -ate > ite > hypo... ite
(-O) (-O) (-O)

adding H: -hydrogen
2H: -dihydrogen

Acids: adding H^+

Anions who end ide: hydro- .... -ic acid

Cl^- Chloride $+H^+$: HCl Hydrochloric acid

Anions who end ate / ite:

-ic | -ous

H^+ + ClO_3^- chlorate → $HClO_3$ chloric acid
H^+ + ClO_2^- chlorite → $HClO_2$ chlorous acid

Amount of Atoms

1	mono	
2	di	Cl_2O dichloride monoxide
3	tri	NF_3 Nitrogen trifluoride
4	tetra	
10	deca	P_4S_{10} Tetraphosphorous Decasulfide

Stochiometry

$2H_2 + O_2 \rightleftharpoons 2H_2O$

Compound Element molecule

Limiting reactant

table with $n = \frac{m}{M_W}$

Gases

$$p = \frac{F}{A} = \left[\frac{N}{m^2}\right] = [Pa] = \left[\frac{kg}{ms^2}\right]$$

$$\rho = \frac{mass}{Volume} = \frac{p}{RT} M_W \Leftrightarrow M_W = \frac{\rho RT}{p}$$

(for gases only)

p_{atm} = 760 mmHg (Barometer)
= 760 torr
= 1,01325 bar
= 101,325 kPa
= 101325 Pa

$\frac{\sim Pa}{101325\,Pa/atm} = atm$

Manometer: $p_{gas} = p_{atm} + \rho g h$

$p_l = \rho g h$

$$pV = nRT$$

p = Pa
V =
R = 8,314

$$R = 8{,}314\,\frac{J}{mol\,K} = \frac{m^3 Pa}{mol\,K} = \frac{L \cdot Bar}{mol\,K} = \frac{L \cdot kPa}{mol\,K}$$

units!

$P_i V_i = P_f V_f$ if n, V const: $\frac{P_i}{T_i} = \frac{P_f}{T_f}$ $\boxed{\frac{P_i V_i}{T_i} = \frac{P_f V_f}{T_f}}$

$\frac{1}{2} m v^2$

Partial Pressure: $P_{tot} = \overbrace{\frac{n_1 RT}{V}}^{P_1} + \overbrace{\frac{n_2 RT}{V}}^{P_2} + \dots + \overbrace{\frac{n_i RT}{V}}^{P_i}$ Daltons law of partial Pressures

mole fraction $P_i = \frac{n_i}{n_{tot}} \cdot P_{tot} = X_i \equiv \frac{n_i}{n_{tot}}$ = mole fraction

$v_{\varnothing} = v_{rms} = \sqrt{\frac{3RT}{Mw}}$ (Velocity root mean squared) = average molecular velocity, depends on T & weight

van der Waals equation for non-idealities

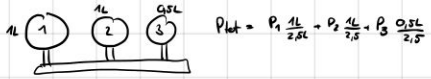
$$\left(P + \frac{n^2 a}{V^2}\right)\left(V - nb\right) = nRT$$

$\frac{n^2 a}{V^2}$: correction for Interactions; nb: corrections for missing Volume

mol % = pressure %

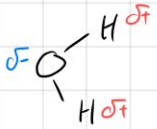
ex: 0,0004 mol % CO_2 →

$P_{CO_2} = 0{,}0004\ atm = 0{,}0405\ kPa$



Prefix	Abbreviation	Meaning	Example	
Peta	P	10^{15}	1 petawatt (PW)	$= 1 \times 10^{15}$ watts[a]
Tera	T	10^{12}	1 terawatt (TW)	$= 1 \times 10^{12}$ watts
Giga	G	10^{9}	1 gigawatt (GW)	$= 1 \times 10^{9}$ watts
Mega	M	10^{6}	1 megawatt (MW)	$= 1 \times 10^{6}$ watts
Kilo	k	10^{3}	1 kilowatt (kW)	$= 1 \times 10^{3}$ watts
Deci	d	10^{-1}	1 deciwatt (dW)	$= 1 \times 10^{-1}$ watt
Centi	c	10^{-2}	1 centiwatt (cW)	$= 1 \times 10^{-2}$ watt
Milli	m	10^{-3}	1 milliwatt (mW)	$= 1 \times 10^{-3}$ watt
Micro	μ[b]	10^{-6}	1 microwatt (μW)	$= 1 \times 10^{-6}$ watt
Nano	n	10^{-9}	1 nanowatt (nW)	$= 1 \times 10^{-9}$ watt
Pico	p	10^{-12}	1 picowatt (pW)	$= 1 \times 10^{-12}$ watt
Femto	f	10^{-15}	1 femtowatt (fW)	$= 1 \times 10^{-15}$ watt
Atto	a	10^{-18}	1 attowatt (aW)	$= 1 \times 10^{-18}$ watt
Zepto	z	10^{-21}	1 zeptowatt (zW)	$= 1 \times 10^{-21}$ watt

Electrolytes



Nonelectrolytes

Precipitation Reactions → solubility chart!

$Ax + By \rightarrow Ay + Bx$

$Ag^+(aq) + NO_3^-(aq) + K^+(aq) + Cl^-(aq) \rightarrow AgCl(s) + K^+(aq) + NO_3^-(aq)$

⇒ K^+ and NO_3^- are on both sides → spectator ions

→ net reaction $Ag^+ + Cl^- \rightarrow AgCl(s)$

$n = \frac{m}{Mw}$

Significant Figures

Any non 0 digit	64,35	4
Any 0 between Digits	740,0035	7
0 on beginning of Number are not sig.	0,00035	2
0 at end are sig. if there is a dec. point	7,600	4

Rounding: if multiplication: to the least sig figs

$1\ kcal = 4{,}184 \cdot 10^3\ J$

Solubility

Soluble Ionic Compounds		Important Exceptions
Compounds containing	NO_3^-	None
	CH_3COO^-	None
	Cl^-	Compounds of Ag^+, Hg_2^{2+}, and Pb^{2+}
	Br^-	Compounds of Ag^+, Hg_2^{2+}, and Pb^{2+}
	I^-	Compounds of Ag^+, Hg_2^{2+}, and Pb^{2+}
	SO_4^{2-}	Compounds of Sr^{2+}, Ba^{2+}, Hg_2^{2+}, and Pb^{2+}
Insoluble Ionic Compounds		**Important Exceptions**
Compounds containing	S^{2-}	Compounds of NH_4^+, the alkali metal cations, Ca^{2+}, Sr^{2+}, and Ba^{2+}
	CO_3^{2-}	Compounds of NH_4^+ and the alkali metal cations
	PO_4^{3-}	Compounds of NH_4^+ and the alkali metal cations
	OH^-	Compounds of NH_4^+, the alkali metal cations, Ca^{2+}, Sr^{2+}, and Ba^{2+}

STP

(for thermodynamic tables)

Thermodynamics → STP = 25°/298K

Pressure → STP = 0°/273K

($PV = nRT$)

Thermodynamics I

three Laws

I Energy is conserved

II Entropy S (= Disorder) is always increasing

III solid at 0°K (T_{abs}) has ΔS entropy of zero.

ΔE if pos: system has gained E (from surroundings)
if neg: system has lost E (to surr)

System types
- **open** matter & energy can be exchanged w/ surroundg
- **closed** energy can be exchanged (but not matter) with surr.
- **Isolated** nothing can be exchaged

Endotherm $q>0$ heat absorbed by sys

Exotherm $q<0$ heat released by sys

Internal Energy E

$$\Delta E = q + w$$

$\Delta E = E_f - E_i$

q = **heat** added to sys

w = **work** done on sys

$w>0$ → work done on sys
$w<0$ → work done by sys

$w = F \cdot s$

Enthalpy H $\left[\frac{kJ}{mol}\right]$

Pressure-Volume work $P\Delta V = RT\Delta n$

$$H = E + PV$$

$$w = -P\Delta V = -RT\Delta n$$ (for Gases)

$$\Delta H = \Delta E + P\Delta V = q_p$$ @ P = konst

q_p = heat added

change in Enthalpy is equal to heat q_p @ const P

$\Delta H - \Delta E = P\Delta V \xrightarrow{\lim V \to 0} \Delta H \approx \Delta E$

$\Delta H_{rxn} = \Delta H_f - \Delta H_i = \Delta H_{products} - \Delta H_{reactants}$

Heat capacity

heat flow required to increase n moles / m grams of substance by ΔT

c_s: How much heat to raise 1g by 1K

molar $q = n \cdot c_m \cdot \Delta T$ $\left[\frac{J}{mol\,K}\right]$ für 1mol

specific $q = m \cdot c_s \cdot \Delta T$ $\left[\frac{J}{g\,K}\right]$ für 18g zb.

if ΔT pos: $\Rightarrow T_f > T_i \Rightarrow q>0$

Hess Law

$\Delta H_{rxn} = \Delta H_1 + \Delta H_2 + \Delta H_3 + \cdots + \Delta H_i$

Enthalpy of formation H_f° (STP)

Enthalpy required to create 1mol of substance from Elements (at STP)

Using ΔH_f° to determine ΔH_{rxn}

$C_3H_8(g) + 5O_2(g) \longrightarrow 3CO_2(g) + 4H_2O(l)$

uniform reactant / form products:

$C_3H_8(g) \rightarrow 3C + 4H_2(g)$	$-\Delta H_f^\circ [C_3H_8] = -(-104\,kJ)$
$C + O_2(g) \rightarrow CO_2(g)$	$\Delta H_f [CO_2(g)] = -394\,kJ$
$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$	$\Delta H_f [H_2O(l)] = -286\,kJ$

$\Delta H_{rxn} = -\Delta H_f [C_3H_8] - 5\,\Delta H_f [O_2] + 3\,\Delta H_f [CO_2(g)] + 4\,\Delta H_f [H_2O(l)]$

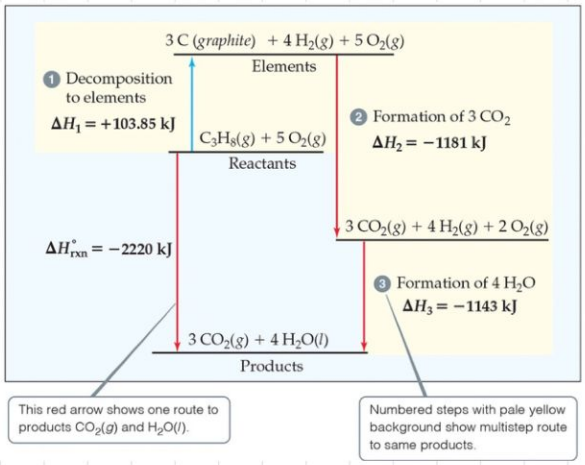
$= -2220\,\frac{kJ}{mol}$

$$\Delta H^\circ_{rxn} = \sum_{products} \beta_i \cdot \Delta H_f^\circ - \sum_{reactants} \alpha_i \cdot \Delta H_f^\circ$$

$\Delta H_{rx} = \sum$ Bonds Broken − $\sum$ Bonds Formed

ΔH_f Products − ΔH_f reactants

α, β etc = stoch. coeff.



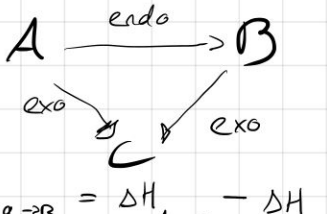
Spontaneous Process

if Process spontaneous in one direction → nonspontaneous in reverse

Nonspontaneous ≠ impossible (possible if we add energy from outside)

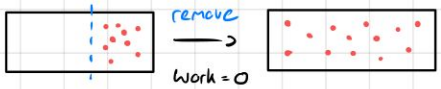
Reversible Process = can be reversed without changing surroundings (T, P, elements etc)

Irreversible Process = surroundings are changed



$\Delta H_{A \to B} = \Delta H_{A \to C} - \Delta H_{B \to C}$

isothermal expansion:



spontaneous & irreversible → requires work to undo

All spontaneous processes are irreversible (→ to return to initial state requires work)

Entropy S $\left[\frac{kJ}{mol}\right]$ (= chaos)

measures tendency for E to spread, disorder } in system
number of microstates, randomness

$$\Delta S^\circ_{rxn} = \sum \beta_i \cdot S^\circ(\text{products}) - \sum \alpha_i \cdot S^\circ(\text{reactants})$$

(α, β stoichiometric coeff.)

for isothermal rxn at P = const:

$$\Delta S = \frac{q_{rev}}{T}$$

q_{rev} = heat flow for reversible process $\approx \Delta H$
T = absolute T

example: ΔS for ice melting:

$\Delta H^{H_2O}_{fus}$ (= melting) $= 6 \cdot 10^3 \frac{J}{mol}$

$q_{rev} = \Delta H_{fus}$ $T_{surr} = 0°C + \delta T$ (δT so small bigger)

$\Delta S = \frac{6 \cdot 10^3}{273\,K} = 22 \frac{J}{mol\,K}$

$$\Delta S_{universe} = \Delta S_{sys} + \Delta S_{surr}$$

$$\Delta S_{surr} = \frac{-\Delta H^\circ_{rxn}}{T}$$

(II) for any spontaneous Process S_{univ} always Increases

(III) solid, $T = 0\,K \rightarrow S = 0$ ($W = 1$)

Reversible: $\Delta S_{univ} = 0$
Irreversible $\Delta S_{univ} > 0$

(Spontan: $\Delta S_{univ} > 0$)

Boltzmann

$$S = k_B \ln(W)$$

k_B = boltzmannsconst. $= 1,38 \cdot 10^{-23} \frac{J}{K}$
W = # of microstates

$$\Delta S = k_B \ln\left(\frac{W_{final}}{W_{initial}}\right)$$

S goes up with V↗, T↗, n↗, melting, vap
⇒ everything that increases chaos

also R is related to k_B: $R = k_B \cdot N_A$

$k_B = \frac{\text{Energy}}{\text{molecule K}}$ $R = \frac{\text{Energy}}{\text{mol K}}$

Gibbs free Energy G

max amount of work we can extract from process is ΔG! (P, T const)

$$\Delta G^\circ = \sum \beta_i \Delta G^\circ_f(\text{products}) - \sum \alpha_i \Delta G^\circ_f(\text{reactants})$$

(α, β stoichiometric coeff.)

$$\Delta G = \Delta H - T \cdot \Delta S$$

$\Delta G = 0$ for natural elements — T is important

ΔG pos → nonspontan
ΔG neg → {spontan; irreversible ← ΔS_{univ} pos}
$\Delta G = 0$ → reversible ← $\Delta S_{univ} = 0$

Finally, let's note that if ΔG is giving us critical information about a process (if it is spontaneous, how much work we can extract, etc.), we should consider the influence of temperature

Importance of T

$\Delta G = \Delta H - T\Delta S$ Spontaneous if $\Delta G < 0$

Example	ΔH	ΔS	$-T\Delta S$	ΔG	Temperature dependence
$2O_3(g) \rightarrow 3O_2(g)$	−	+	−	−	Spontaneous at all T
$3O_2(g) \rightarrow 2O_3(g)$	+ endo	order	+	+	nonspontaneous at all T
$H_2O(l) \rightarrow H_2O(s)$	− exo	disorder	+	+ or −	Spontaneous at low T
$H_2O(s) \rightarrow H_2O(l)$	+ endo	disorder	−	+ or −	Spontaneous at high T

(Spontan: favored by both enthalp and entrop; Favored by neither; Favored only by enthalpy; Favored only by entropy)

order

This term becomes larger in magnitude as T increases.
Explains why ice melts above 0°C: $|-T\Delta S|$ becomes larger than $|\Delta H|$

	$\Delta H < 0$	$\Delta H > 0$
$\Delta S > 0$	Spontaneous at all T ($\Delta G < 0$)	Spontaneous at high T (when $T\Delta S$ is large)
$\Delta S < 0$	Spontaneous at low T (when $T\Delta S$ is small)	Non-spontaneous at all T ($\Delta G > 0$)

Endo	Exo
q > 0	q < 0
$\Delta H > 0$	$\Delta H < 0$

Gibbs & equilibrium

ΔG°_f (at STP) = Free energy to form substance from elements (formation)

ΔG° = Free energy change for any rxn

$\Delta G < 0$, $K > 1$ equil. → products
$\Delta G > 0$, $K < 1$ equil. → reactants

rxn spontaneous: $\Delta G < 0$
at equilibrium $\Delta G = 0$

$$\Delta G = \Delta G^\circ + RT \ln(Q)$$

at equilibrium

T = absolute T
R = Gas const.
Q = rxn quotient
Q = K

Gibbs redox

$$\Delta G^\circ = -n F E^\circ_{cell}$$

$\Delta G^\circ = \frac{J}{mol}$ of rxn
F = Faraday's constant $= 96{,}485 \frac{C}{mol\ e^-}$
n = positive, unitless number moles of e^- transferred in balanced cell rxn

$$\Delta G^\circ = -RT \ln(K) \Leftrightarrow K = \exp\left(\frac{-\Delta G}{RT}\right)$$

lecture 6

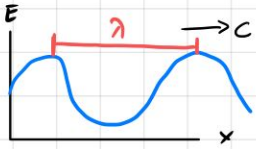
Light

$$\nu = \frac{c}{\lambda}$$

c = speed of light $= 2{,}998 \cdot 10^8 \frac{m}{s}$

λ = wavelength

ν = frequency



Colour varies with λ:

430 nm blue

530 nm green $\quad nm = 10^{-9} m$

630 nm red

$$E_{Light} = h \cdot \nu = \frac{hc}{\lambda}$$

h = Planks konstant $= 6{,}62607 \cdot 10^{-34} \frac{m^2 kg}{s} = \frac{J}{s}$

Energy ⇒ photons

⇒ thus the shorter wavelength → the higher frequency the more energy light has

$E \nearrow \Rightarrow \lambda \searrow\!\searrow$, $\nu \nearrow$

E blue > green > red

History of hydrogen (H_2)

$$E_H = \frac{hc}{\lambda} = hcR_H\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

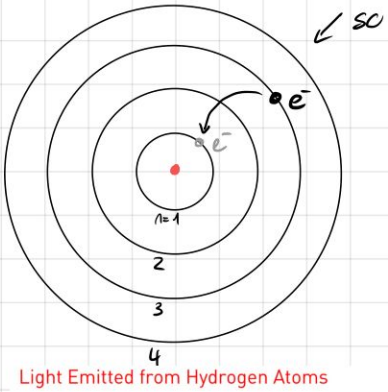
R_H = Rydberg constant $= 1{,}097 \cdot 10^7 \frac{1}{m}$

n_1, n_2 = Integers

$n_2 > n_1 > 0$

$$\lambda = \frac{hc}{E} = \frac{\cancel{hc}}{\cancel{hc} R_H\left(\frac{1}{n_1} - \frac{1}{n_2}\right)} = \frac{1}{R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)}$$

Bohr Model

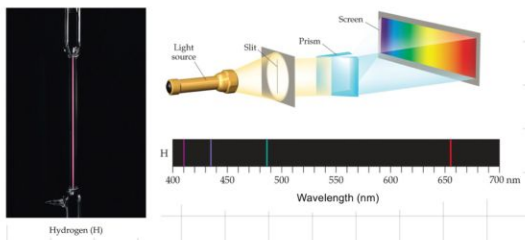


- electron can move between orbits
- electron has specific "allowed Energies"
  - ↳ but this requires absorption or emission of photon $E_{ph} = h\nu$

Ex: $E_3 \to E_1$, with $E_3 > E_1 \Rightarrow \Delta E_{3\to1} = -E_{photon}$

Emitting E neg
Absorbing E pos (?) losing energy

Light Emitted from Hydrogen Atoms



Hydrogen (H)

Orbital Energies

$$E_n = -\frac{hcR_H}{n^2}$$

$h = 6{,}62607 \cdot 10^{-34} J \cdot s = \frac{m^2 kg}{s}$

$R_H = 1{,}097 \cdot 10^7 \frac{1}{m}$

$n = 1, 2, 3, 4$ Energy levels specified by "n"

- Energy levels are negative bc, electron lowers its energy (energetisch günstiger) by getting closer to the nucleus (+ & - attract)
- Energy of emitted Light is negativ
- The further away from the nucleus are, the more energy they have

lec. 6

Quantum Mechanics

Uncertainty Principle $= \Delta x \cdot \Delta p \geq \frac{h}{4\pi}$

how well we know
Δx position
Δp momentum

4 Quantum Numbers

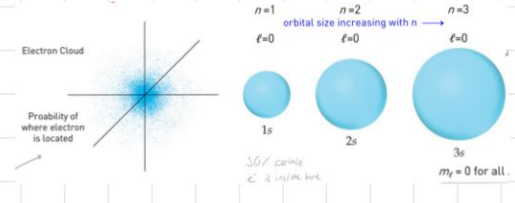
→ n = Shell = size of orbital

$n = 1, 2, 3, 4, 5, 6, 7, 8$

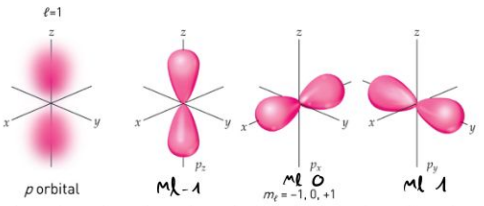
→ l = subshell = shape of orbital $\quad l = 0, 1, 2, 3, \ldots n-1$

l =	0	1	2	3
	s	p	d	f

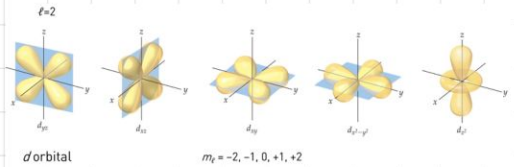
s orbitals $l = 0$



p Orbitals $l = 1$



d Orbitals $l = 2$



f orbitals $M_l = \{-3, +3\}$

→ M_l = orientation → depends on l

$M_l = \{-l \text{ bis } l\}$

→ M_s = spin up or down $m_s = \pm \frac{1}{2}$ ↑↓

how electrons are in orbital.

Energy depends on n → 2s, 2p / 3s, 3d, 3p ⇒ have the same Energy

Fill from bottom up, then go again, two e⁻ in one box (spin)

No e⁻ can have same set of n, l, M_l, M_s

l orbital

↑↓ ↑↓ ↑↓ ↑↓ ↑ ↑ ↑ ↑

Pauli Exclusion Principle: Not two electrons can have the same set of Quantum Numbers

$S = 2$ $p = 6$ $d = 10$ $f = 14$

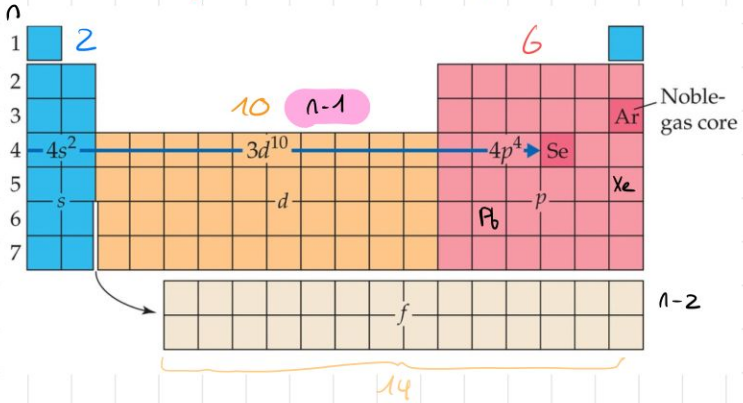


TABLE 6.3 Electron Configurations of Several Lighter Elements

Element	Total Electrons	Orbital Diagram 1s	2s	2p	3s	Electron Configuration
Li	3	↑↓	↑			$1s^2 2s^1$
Be	4	↑↓	↑↓			$1s^2 2s^2$
B	5	↑↓	↑↓	↑		$1s^2 2s^2 2p^1$
C	6	↑↓	↑↓	↑ ↑		$1s^2 2s^2 2p^2$
N	7	↑↓	↑↓	↑ ↑ ↑		$1s^2 2s^2 2p^3$
Ne	10	↑↓	↑↓	↑↓ ↑↓ ↑↓		$1s^2 2s^2 2p^6$
Na	11	↑↓	↑↓	↑↓ ↑↓ ↑↓	↑	$1s^2 2s^2 2p^6 3s^1$

Box = orbital

⇒ Hunds rule: fill up orbital box first, then 2nd spin

place electron spins parallel first in whole subshell
then add second spin

Elektronenkonfiguration

Edelgas von Reihe davor nehmen, dann Reihe durchgehen bis zu Element, und jeweils Schale, orbital ^ schreiben, Schale des d orbitale -1, & Schale f orbitale -2, (warum? Ka)

Bsp: Se [Ar] $4s^2 3d^{10} 4p^4$

Bsp: Pb [Xe] $6s^2 4f^{14} 5d^{10} 6p^2$ (n-2, n-1)

Atomic size



Bonding atomic radius = $\frac{d}{2}$

Atomic radius tend to increase down in column PSE (n shell is getting bigger) & decrease across Period ⇒ because Zeff increases.

Ionisation Energy

Energy to remove e⁻

$Na \rightarrow Na^+ + 1e^-$ (I_1) first

$Na^+ \rightarrow Na^{2+} + 1e^-$ (I_2) second Ionisation Energies

Trends $I_2 > I_1$

I increases from left to right PSE

I decreases down column PSE

PSE

I > 0	add Energy to system to remove e⁻
EA < 0	system lowers Energy by accepting e⁻ and making anion

Electron Affinity

eigentlich negative Zahl

Energy change when add extra e⁻ to Atom

Typically Exothermic: EA < 0

Trends: Halogens EA << 0, Noble gases EA > 0 (full shell)

Screening e⁻ Attraction

Z_{eff} increases across row in PSE

Zeff = attraction Ve to Nucleus(?)

Ionic Radius

Estimate radius from Bond lengths in ionic solids

Cations ⊕ smaller radius
Anions ⊖ bigger radius than parent neutral atom

Trends: Ions with same charge → bigger down group (column)
Ions with same # e⁻ → smaller with increasing Zeff (row)

Bonds

Octet rule: Atom wants 8 Ve^- — :Xe:

1) Ionic

Na^+ Cl^- — electrostatic

Strength qualified by Energy to separate two

$NaCl \xrightarrow{\Delta H_{lat}} Na^+ + Cl^- \quad \Delta H + 788 \frac{kJ}{mol}$

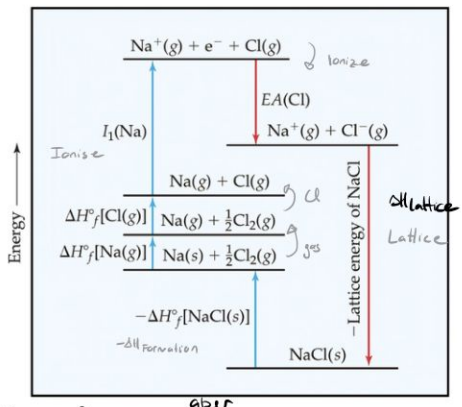
reverse $Na^+ + Cl^- \longrightarrow NaCl \quad \Delta H - 788 \frac{kJ}{mol}$

Lattice Energy: energy to separate two ions ∞ distance apart

Ionisation Energy

$(Atomic\#-1)^2 \cdot 13{,}6 = V$

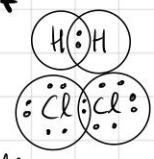
Calculating Lattice Energy



alles aufsummieren, $(-H_f, -E_{lat}), = 0$, nach lat aufl.

2) Covalent

H_2O — sharing Ve^-



Each atom has Noble gas config

Multiple Bonds

$O=O$ (->=>≡)

$N \equiv N$

Trend: Bond length decreases with # shared e^-

Electronegativity

How strong an Atom pulls e^-

Use to estimate Bond polarity $\delta+ \; \delta-$

Trend: Increases across row, decreases down column — PSE ↑ incr.

Electronegative Atoms
- Large negative EA
- Large Ionisation Energy

(want Electrons and resist giving e)

ΔElectroneg

$\Delta E_N > 2$ Ionic Bond

$\Delta E_N > 0{,}4$ Polar

$\Delta E_N < 0{,}4$ Apolar

As Ox Number of Metal increases ⇒ More covalent

Dipole Moments

Dipole moment $\vec{\mu}$ — $\delta+ H - F \; \delta-$ — Vector from neg to pos

$O=C=O$ — apolar bc linear symmetry

sum $\mu_2 + \mu_1 = 0$

$|\mu| = Q \cdot r \quad [D]$

Debye $\equiv 3{,}34 \times 10^{-30}$ Cm

r = distance

Q = Charge

μ = Magnitude

3) Metallic Bonds

Fe

Ve^- shared with whole solid

Lewis Structures

1) Sum of all Ve^-
2) Distribute, complete Octet for Nonentral Atoms first
3) Multiple Bonds if Central Atom doesn't have Octet

exceptions to octet rule: odd e^- $\bar{N}=\dot{O}$, $\bar{N}=\dot{C}$
less than 8 e^- on central atom
more than 8 e^- on central atom: hypervalent

Alternative Lewis Structures:

	$\ddot{O}=C=\ddot{O}$			$:\ddot{O}-C\equiv O:$		
Valence electrons:	6	4	6	6	4	6
−(Electrons assigned to atom):	6	4	6	7	4	5
Formal charge:	0	0	0	−1	0	+1

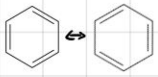
Formal Charge: Charges Atoms would have if Bond e shared equal (or: Valence e^- − e^- assigned to atom)
$\#Ve^- - \frac{1}{2}$ Bonding e^- − lone e^-

Ox Number: Charges Atoms would have if Bond Ionic, Shared e^- given to Atom more EN

Partial Charge: $\delta+ \; \delta-$

Resonance Structures

Multiple possible Lewis Structures

$\langle O=\bar{O}-\bar{O}| \leftrightarrow |\bar{O}-\bar{O}=O\rangle$ or 

$$\Delta H_{rxn} = \sum \text{Bonds Broken} - \sum \text{Bonds Formed}$$

Shapes: VSEPR (valence shell e^--pair repulsion)

distinguish between:

Electron domain Geometry: Includes both bonding and non bonding pairs (lone pairs)

VS

Molecular Geometry: Includes just atoms

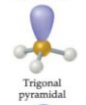
Remember:

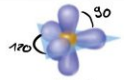
- Lone pairs will take more space than bonds



- double / triple Bonds take more space than single bonds



Number of Electron Domains	Electron-Domain Geometry	Bonding Domains	Nonbonding Domains	Molecular Geometry	Example
2	 Linear	2	0	 Linear	
3	 Trigonal planar	3	0	 Trigonal planar	
		2	1	 Bent	
4	 Tetrahedral	4	0	 Tetrahedral	
		3	1	 Trigonal pyramidal	
		2	2	 Bent	

Number of Electron Domains	Electron-Domain Geometry	Bonding Domains	Nonbonding Domains	Molecular Geometry	Example
5	 Trigonal bipyramidal	5	0	 Trigonal bipyramidal	PCl_5
		4	1	 Seesaw	SF_4
		3	2	 T-shaped	ClF_3
		2	3	 Linear	XeF_2
6	 Octahedral	6	0	 Octahedral	SF_6
		5	1	 Square pyramidal	BrF_5
		4	2	 Square planar	XeF_4

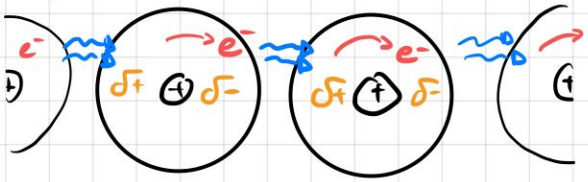
Intermolecular Forces (ZMK) (no Bonds)

E_{int} := Energy of Interaction between Molecules (Anziehung)
E_{kin} := Kinetic Energy of molecules (Bewegung)

E_{int} set by RT, depends from separation Distance d
If Temp increases → $|E_{kin}|$ ↗, $|E_{int}|$ ↘

3 Types of Interactions

(I) Induced Dipole / Dispersion (always present)



→ always possible
very attractive, but only on short distances.

Temporary Dipol

(I), (II) VdWK

very short Range, but Menge makes the difference

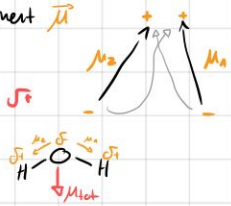


→ All Atoms, always present
→ depends on polarisability

(II) Dipol-Dipol Interaction (present for polar molecules) (permanent), strongest

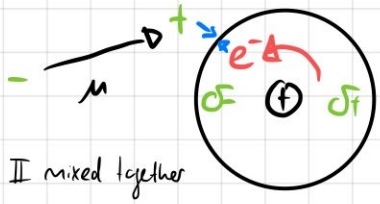
Permanent Dipole is needed for this Interaction
Polar bonds → Molecule with dipole moment $\vec{\mu}$
Polar molecules have $\mu_{tot} \neq 0$

$|\vec{\mu}| = q \cdot d$ ($\oplus$ q^+ ← $\vec{\mu}$ $\ominus$ q^-)



vector sum of all bond Dipole Moments

(II.5) Dipole-Induced Dipole

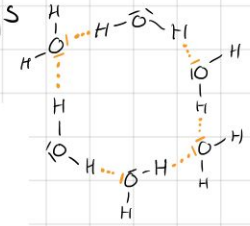


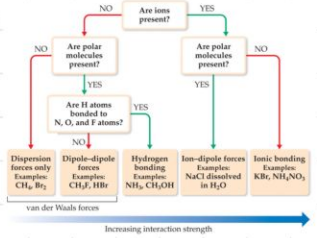
I and II mixed together

(III) Hydrogen Bonding / H-Brücken

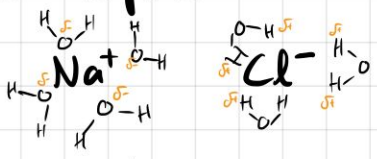
only F, O, N + H Bonds (they are very electronegative
H can attatch to other lone e⁻ pairs of F, O, N Atoms
Active Halbbrücke wenn Molekül mit anderen gleiche Sorte H Brücken machen kann
Passive Halbbrücke: wenn es eine andere Molekül sorte braucht

Thats why Liquid water is more dense than ice, because in (frozen)? state it forms rings





(IV) Ion-Dipole



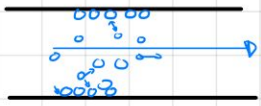
→ solubility

Strength

Bonds > Ion-Dipole > H-Bonds > Dipole-Dipole > Induced Dipole

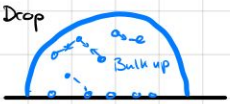
Properties of Liquids

Viscosity $\left[\frac{kg}{ms}\right]$



layers of fluid in the middle go fasta (they eat pasta)
the outer one stick to wall (Velvik)

Surface Tension $\left[\frac{J}{m^2}\right]$



E_{int} is strong: Liquid balls up
Intermolecular Attraction

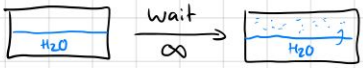


Maximizes Molecular E_{int}
if liquid high surface tension $\rightarrow$ high $\frac{bulk}{interface}$

Vapor Pressure

affected by P & T

T = Konst: $(P_{vap} = P_{sat})$

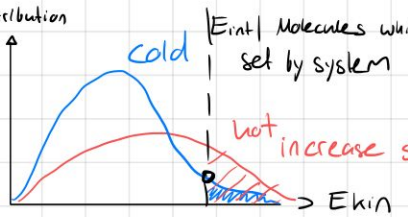


T ≠ Konst:

T ↑ E_{kin} ↗ E_{int} ↘ (less strong), more leave H_2O

T ↓ E_{kin} ↘ E_{int} ↗, gas condenses

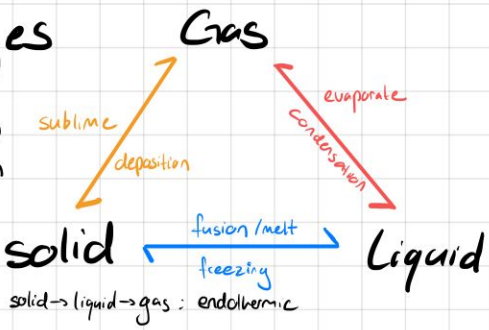
E_{kin} vs E_{int}



blue Area: Molecules that have enough Energy to evaporate at low T

red Area: Molecules enough Energy to evaporate at higher Temp.

Phase changes



ΔH_{fus} = heat of fusion
ΔH_{vap} = heat of vaporisation
ΔH_{sub} = heat of sublimation
$\Delta H_{sub} = \Delta H_{vap} + \Delta H_{fus}$
$\Delta H_{vap} > \Delta H_{fus}$

solid → liquid → gas : endothermic

Heating Curves

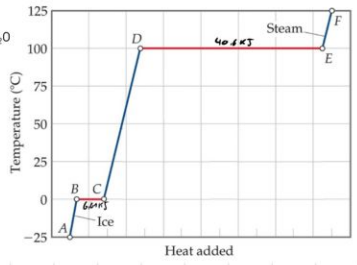
Ex: How much energy is required to heat 1 mol H_2O from −25.0°C to 125°C at 101.3 kPa?
A → F

Molar heat capacities:
Ice = 38.1 J mol^{-1} K^{-1}
Liquid water = 75.3 J mol^{-1} K^{-1}
Steam = 37.5 J mol^{-1} K^{-1}

Molar Enthalpies
ΔH_{fus} = 6.01 kJ mol^{-1}
ΔH_{vap} = 40.6 kJ mol^{-1}

Enthalpy Changes:
A to B = 0.953 kJ heat
B to C = 6.01 kJ fusion
C to D = 7.53 kJ heat
D to E = 40.6 kJ vap
E to F = 0.938 kJ heat

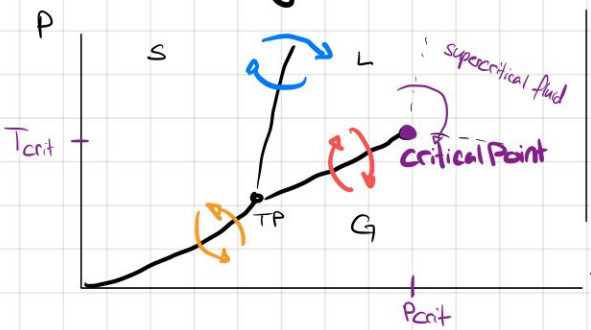
Entire Process
ΔH = 56.0 kJ



$$\ln P_{vap} = \frac{-\Delta H_{vap}}{RT} + C$$

T = Tabsolut
R = 8,314
C =

Phase Diagram



Critical Point: highest T, P where distinct liquid Phase exists
Large Intermolecular Forces
⇒ high T_{crit}

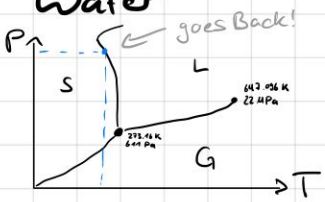
$T_{melting}$ normally increases with increasing P (except H_2O)

$T_{boiling}$ increases with increasing P
↳ boiling when $P_{vap} = P$



equation for water boils at T=25° at vacuum

Water



→ thats why ice scating works! ice scate exerts so much pressure so that the ice beneath the iron turns liquid and you slide on a small film of water!

→ generally if Water behaves strangely
→ first answer: cause of H-Bondings

$$\ln\left(\frac{P_1}{P_2}\right) = \frac{\Delta H}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

lecture 9 Thermodynamics of Solutions

Define: solvent + solute → Solution (→ change in ΔGibbs free Energy)
Lösungsmittel gelöster Stoff → Lösung

Solution spontaneous if $\Delta G_{soln} < 0$

or $\Delta G_{soln} = \Delta H_{soln} - T\Delta S_{soln}$

ΔH depends on Intermolecular Interactions at fixed P

ΔH_{soln}: (i) $(solvent)_n \longrightarrow n \cdot solvent$ $\Delta H_{solvent}$
(ii) $(solute)_m \longrightarrow m \cdot solute$ ΔH_{solute}
(iii) $n \cdot solvent + m \cdot solute \longrightarrow solution$ ΔH_{mix}

$(solvent)_n + (solute)_m \longrightarrow Solution$ ΔH_{soln}

$$\Delta H_{soln} = \Delta H_{solvent} + \Delta H_{solute} + \Delta H_{mix}$$

↑ total Enthalpy change

ΔH_{soln} can be (+) Endotherm or (−) Exotherm and since $\Delta G = \Delta H - T\Delta S$

if $\Delta H_{soln} < 0$

Typically solution forms for any solute concentration

pure solvent ⟷ pure solute

Miscible "mixable"

if $\Delta H_{soln} > 0$

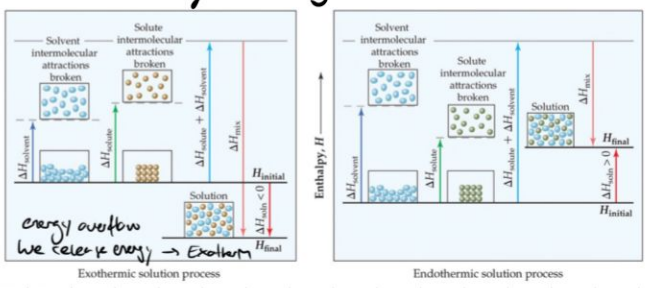
$-T\Delta S_{soln}$ must be sufficiently neg to compensate for pos ΔH, to make $\Delta G_{soln} < 0$

Otherwise solute will not dissolve

Process will depend on T, more likely at increasing T.

But if ΔH is very pos. the ΔH may not be enough to compensate → will not dissolve → insoluble

Enthalpy changes during solution



Solubility S $[\frac{g}{L}]$ max dilute = wassrig

Max. Amount of solute in solvent at given T

⇒ **Saturated Soln** = solute + solvent ⇌ solution (dissolve / crystallize)

↳ Dynamic equilibrium above solubility concentration

supersaturated soln → contains more than max amount of capable at given T

(ii) Pressure

Solub. of (g) in a (l) depends on P_g

$$S_g = k \cdot P_g$$

Henrys Law

S_g = gas solubility – molar concentration?

P_g = gas partial pressure : P. volatilität

k = Henrys Law konstant

↳ need to calculate each time for Stoff and Temp fixed

(iii) Temp

Ion solubility in H_2O ↗ if T ↗ $\Delta S_{soln} > 0$

Gas solubility in H_2O ↘ if T ↗ $\Delta S_{soln} < 0$

Factors affecting solubility

(i) intermolecular

strong solute – solvent interaction favour

Solids: Ions dissolve more easily in polar solvents

Liquids: polar liquids diss. easily in polar liquids

non-polar liquids are immiscible with polar liq.

General: if similar Intermolec. Interactions → soluble

Concentration

X	mole fraction = $\frac{\text{moles solute}}{\text{total moles}}$	useful for gas mix	mass % = $\frac{\text{mass solute}}{\text{total mass}} \times 100$
M	Molarity = $\frac{\text{moles solute}}{\text{Litres solution}}$ = c	Depends on Volume of solution	ppm parts per million = $\frac{\text{mass solute}}{\text{total mass}} \times 10^6$
m	Molality = $\frac{\text{moles solute}}{\text{kg solvent}}$	Depends on mass of solvent → useful if T variable bc kg does not increase with T	ppb parts per billion = $\frac{\text{mass solute}}{\text{total mass}} \times 10^9$

Colligative Properties

solute affects properties of liquids

If depends only on amount of solute → colligative

dilution formula: c = concentrated, d = diluted

$$M_c \cdot V_c = M_d \cdot V_c$$

(A) Boiling Point ↕

$$\Delta T_{bp} = T_{bp}^{soln} - T_{bp}^{pure} = i k_b m$$

m = molality

K_b = molal bp elevation konst.

i = Vant Hoff factor

i = 1 for non-electrolytes.

of Ions produced for electrolytes "ions"

ex: NaCl i = 2, $MgCl_2$ i = 3

Water: if we add NaCl T_{bp} ↗

(B) P_{vap} ↘

Liquid boils when $P_{vap} = P$

bsp. volatile solvent + non-volatile solute

For ideal Soln

$$P_{vap}^{soln} = X_{solvent} \cdot P_{vap}^{pure}$$

X = mole fraction of solvent in solution

Ideal Gas: No Intermolec. Interactions ⇒ $PV = nRT$

vs.

Ideal Soln: All Intermolec. Int identical

$\text{solute}^{int}\text{-solute} = \text{solvent}^{int}\text{-solvent} = \text{solute}^{int}\text{-solvent}$

if solute–solvent < {solv–solv, solu–solu} → $P_{vap}^{soln,actual} > P_{vap}^{soln\,Raoult}$ bigger than ideal case

if solute–solvent > {solv–solv, solu–solu} → $P_{vap}^{soln,actual} < P_{vap}^{soln\,Raoult}$ less than Ideal case

(C) Freezing Point ↘

$$\Delta T_{fp} = T_{fp}^{soln} - T_{fp}^{pure} = -i k_f m$$

m = molality of solute
K_f = molal fp depression konst.
i = Vant Hoff factor = # electrolytes

ex: $K_{f\,H_2O} = 1{,}86\ ^\circ C/m$
equation is for dilute Soln that are ideal
→ At higher concentration typically nonideal

(D) Osmosis

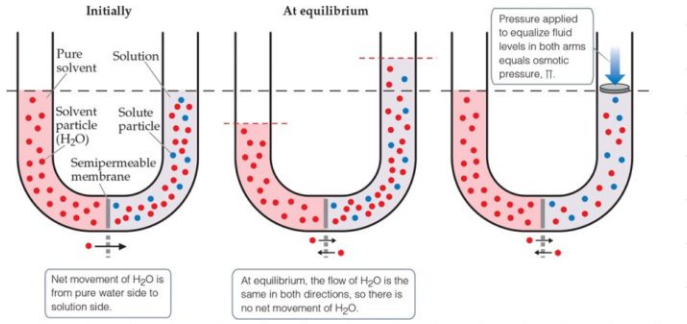
semi-permeable membrane only passes solvent, not solu
→ Osmotic flow: tries to equalize concentrations
→ Osmotic pressure: p needed to counteract osm flow
wants equilibrium

Osmotic Pressure

$$\Pi = iMRT$$

$\Pi V = inRT \mid n = MV$
M = Molarity
i = Vanthoff factor # electrolytes

Can use Π to determine M_w
$\Pi = iMRT \mid M = \frac{n}{V[L]}$
$\Rightarrow \frac{mass}{mole} = M_w$



Two Soln
$\Pi = \Pi$ isotonic
$\Pi > \Pi_{ref}$ hypertonic
$\Pi < \Pi_{ref}$ hypotonic

Lecture 10

Reaction Rate

$$\text{rxn rate} = \frac{\text{change of concentration [C]}}{\Delta t} \quad \left[\frac{M}{s}\right]$$

$\frac{\text{Molarity}}{\text{time}}$ or $\frac{\text{Moles}}{L \cdot s}$
rxn rates always positive

example: rxn rate = rate of appearance = rate of dissapearance

$$\text{rxn rate} = \frac{\Delta[B]}{\Delta t} = \frac{-\Delta[A]}{\Delta t}$$

[C] concentration in Molarity

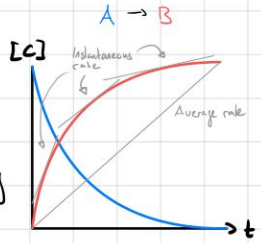
Average rxn rate

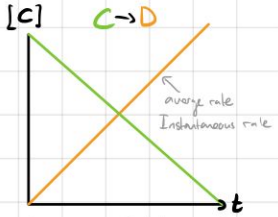
$$\frac{\Delta[B]}{\Delta t} = \frac{[B]_{t_2} - [B]_{t_1}}{t_2 - t_1}$$

Instantaneous rxn rate

$$\frac{\Delta[B]}{\Delta t}\ \Delta t \to 0 = \frac{d[B]}{dt}$$

Ableitung





here: average = Instantaneous rate
→ here rxn rate does not depend on [C]!
→ Block Box Cookie Factory Analogy only one Machine

Rate = Instantaneous Rate at given t, $\left[\frac{M}{s}\right]$

Initial rate = Instantaneous Rate at t = 0

$$\alpha A + \beta B \longrightarrow \gamma C + \delta D$$

$\alpha, \beta, \gamma, \delta$ stoichiometric Coeff.

$$\text{Rate} := -\frac{1}{\alpha}\frac{d[A]}{dt} = -\frac{1}{\beta}\frac{d[B]}{dt} = \frac{1}{\gamma}\frac{d[C]}{dt} = \frac{1}{\delta}\frac{d[D]}{dt}$$

Rate Laws

find out by experiment

$$\text{Rate Law} = k\,[A]^m \cdot [B]^n$$

k = rate law constant
m, n, ... = reaction orders $\in [0, \frac{1}{2}, 1, 2 \ldots]$
! $\Rightarrow \neq \alpha, \beta, \gamma \ldots$

if more reactants $n \cdot [A]^m [B]^n [C]^p \ldots$ only reactants

Overall rxn order = $m + n + p \ldots$ | sum of exponents

Units of k → depend on rxn order!

$$[k] = \frac{M}{s \cdot M^{m+n+p\ldots}} \quad \mid M = \frac{n}{L},\ k = \frac{M}{s}$$

$$\text{units of Rate} = (\text{units of } k) \cdot [C]^{m+n+p\ldots} \longrightarrow$$

rxn order	unit of k
0	$\frac{M}{s}$
$\frac{1}{2}$	$\frac{M^{\frac{1}{2}}}{s}$
1	$\frac{1}{s}$
2	$\frac{1}{M \cdot s}$
3	$\frac{1}{M^2 s}$

How to find out?

Exp	[A] (M)	[B] (M)	Initial rate law (M/s)
1	0,1	0,1	$4 \cdot 10^{-5}$
2	0,1	0,2	$4 \cdot 10^{-5}$
3	0,2	0,1	$16 \cdot 10^{-5}$

$$\frac{\text{Rate 2}}{\text{Rate 1}} = \frac{4 \cdot 10^{-5}}{4 \cdot 10^{-5}} = 1 \quad \mid \text{rate} = k \cdot [A]^m [B]^n$$

$$\Leftrightarrow 1 = \frac{k \cdot 0{,}1^m \cdot 0{,}2^n}{k \cdot 0{,}1^m \cdot 0{,}1^n}$$

$1 = 2^n \rightarrow n = 0$ ✓ | n = rxn order

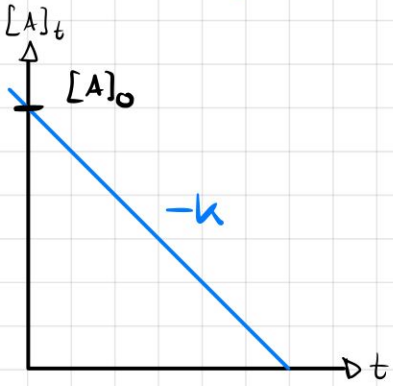
or: $\frac{\text{rate 3}}{\text{rate 1}} = \frac{16 \cdot 10^{-5}}{4 \cdot 10^{-5}} = 4 \quad \mid \text{rate} = k \cdot [A]^m \ldots$

$$4 = \frac{0{,}2^m \cdot 0{,}1^n}{0{,}1^m \cdot 0{,}1^n} = \frac{0{,}2^m}{0{,}1^m} = 2^m = 4 \Rightarrow \underbrace{\text{rxn order } m = 2}$$

$$-\frac{d[A]}{dt} = \text{rate} = k[A]^M \quad | \quad M \in 0,1,2$$

0th order

$M = 0$



$$[A]_t = -kt + [A]_0$$

$A \rightarrow B$: $\text{rate} = -\frac{d[A]}{dt} = \underbrace{k[A]^0}_{=k}$

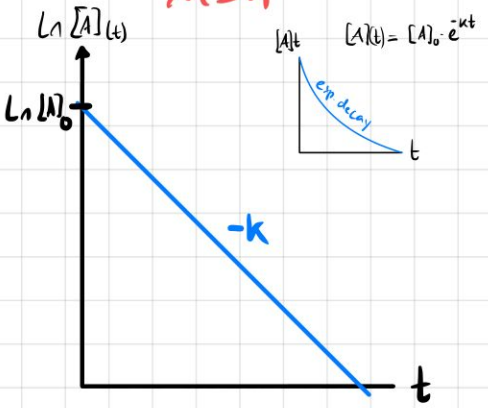
$$t_{\frac{1}{2}} = \frac{[A]_0}{2k}$$

prop. to intial concentration

Like rocket burning fuel at constant rate. $t_{\frac{1}{2}}$ depends on intial amount of fuel $[A]_0$, but not on how much fuel is left $[A]_t$

1st Order

$M = 1$



$$\ln[A]_t = -kt + \ln[A]_0$$

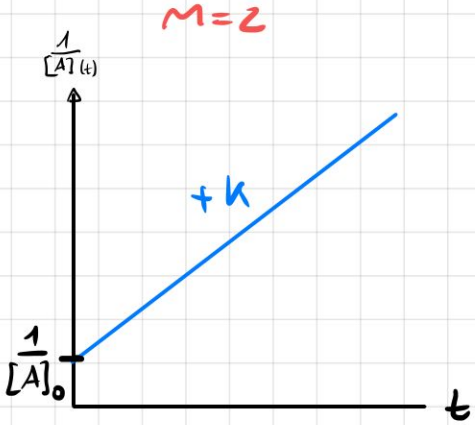
$A \rightarrow B$: $\text{rate} = -\frac{d[A]}{dt} = k[A]^1$

$$t_{\frac{1}{2}} = \frac{\ln(2)}{k} \quad \leftarrow = 0{,}693$$

Does not depend on initial Concentration

2nd order

$M = 2$



$$\frac{1}{[A]_t} = kt + \frac{1}{[A]_0}$$

$A \rightarrow B$ $\text{rate} = -\frac{d[A]}{dt} = k[A]^2$

$$t_{\frac{1}{2}} = \frac{1}{k[A]_0}$$

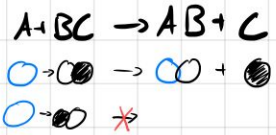
Inversely prop to intial concentration

Ähnlich Molekülstoss?

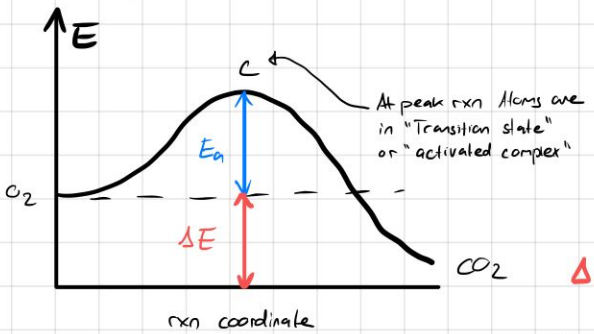
Lecture 11

Chemical rxn: Collision model

rxn rate $\overset{\text{prop.}}{\propto}$ collisions per second | depends on orientation they hit each other



collision frequency depends on concentrations,
at higher T, molecules move faster, therefore collide more often & more forcefully



ΔE neg → exo

ΔE pos → endo

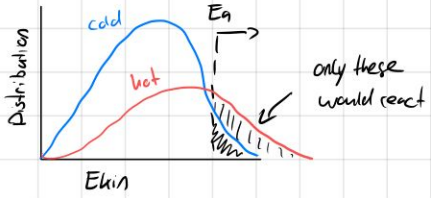
ΔE is irrelevant for Ea

Larger Ea → slower rates

$$Ea = \left[\frac{J}{mol}\right]$$

- Larger Ea → bigger Energetic hill
- Molecules carry energy into rxn via motion
- Molecules have distribution of Ekin

⇒ To react molecules need $E > Ea$



rate = $k[A]^m \cdot [B]^n$ → rate konstant is dependant on T

Arrenhius

$$k(T) = A \cdot \exp\left(-\frac{Ea}{RT}\right)$$

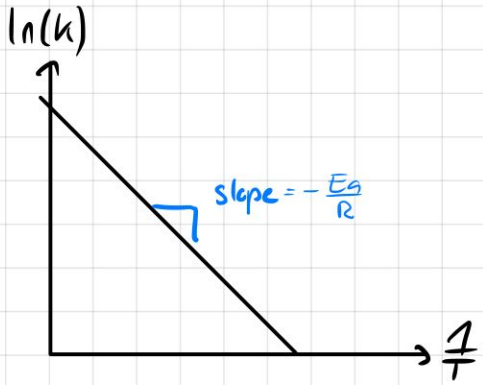
A actually depends on T too, but so little, that we neglect it

$A = \left(\text{collisions per second}\right) \cdot \left(\text{fractions of collisions that are properly oriented to actually react}\right) = \text{const}$

$e^{-\frac{Ea}{RT}} = \left(\text{fractions of molecules that can climb the barrier/mountain}\right)$

$Ea = \left[\frac{J}{mol}\right]$

R = gas constant

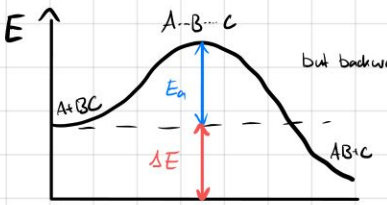


$\Longleftrightarrow$

$$\ln(k) = -\frac{Ea}{RT} + \ln(A) \xrightarrow{\Delta T} \ln\left(\frac{k_1}{k_2}\right) = \frac{Ea}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

Rule of thumb: for typical Ea, T ⇒ rxn rate doubles if $\Delta T = +10°C$

Role of ΔE



A + BC → AB + C

but backwards: A + BC ← AB + C

$Ea^{bwd} = \Delta E + Ea^{fwd}$

→ rxn much slower

Ea is an exp function so small changes in T make big differences

Reaction Mechanisms

describes order bonds break

write rxn als elementary rxn { one thing happens / one bond breaks/forms

each rxn has a molecularity → #moles participating

"1"	unimolecular	A → Products
"2"	Bimolecular	A + A → Products
"3"	Termolecular	A + A + A → Products

Likelyhood increase ↑

↳ less likely to be an elementary rxn

rare for 3 molecules to collide, or even 4

Overall rxn / Global rxn $2H_2 + O_2 \rightarrow 2H_2O$

elementary rxns

$H_2 \rightarrow 2H$

$H + O_2 \rightarrow OH + H$ ← Intermediate

⇒ formed in Step 1

⇒ consumed in Step 2

↳ does not show up in Global rxn

:

if rxn is elementary → m, n, p = Stochiometric coeff.

$\alpha A + \beta B \rightarrow \mu C + \delta D$ rate = $k(T) \cdot [A]^m \cdot [B]^n$ → if elementary $m = \alpha$, $n = \beta$

$k(T) \cdot [A]^\alpha \cdot [B]^\beta$

you cant know if rxn is elementary

↳ chemist discuss

or given an elementary rxn, rxn order depends on Molecularity

Elementary Reactions and Their Rate Laws

P ≡ Products

Molecularity	Elementary reaction	Rate law
Unimolecular	A → P	$k[A]$
Bimolecular	A + A → P	$k[A]^2$
Bimolecular	A + B → P	$k[A][B]$
Termolecular	A + A + A → P	$k[A]^3$
Termolecular	A + A + B → P	$k[A]^2[B]$
Termolecular	A + B + C → P	$k[A][B][C]$

Termolecular less likely than bimolecular

4-molecule collision even rarer

Process to determine rxn Mechanism

→ Experiment
→ Propose idea
→ Check

Example: Proposed Mechanism

slow $2A \xrightarrow{k_1} I + C$ (consumed)

fast $I + B \xrightarrow{k_2} A + D$

} elementary rxn

$2A + I + B \rightarrow A + D + I + C \quad | -A, -I$

$A + B \rightarrow C + D$ } Global rxn

Stochiometric coeff bc. assumed elementary

rate law? $rate_1 = k_1 [A]^2$, $rate_2 = k_2 [I][B]$

↳ system of equations to be solved.

one rxn will be rate limiting

→ one step is so slow, it set rxn order for the Global rxn rate

if $rate_1$ is limiting: $k_1 \ll k_2$ (slow, fast); stochiometric coeff

$rate_{overall} = k_1 \cdot [A]^2 \cdot [B]^0$

→ check with experiments if consistent or not

Catalysts

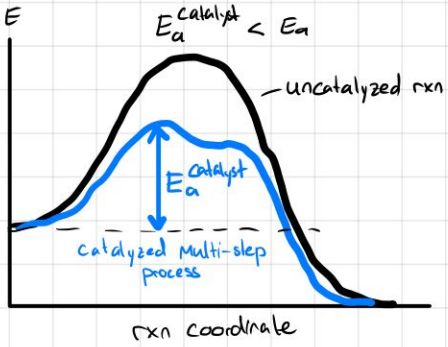
Substances that increase rxn rate

get consumed but reproduced in sub-step

so dont show up in Global rxn

Catalysts affect k

$k = A \cdot \exp\left(-\frac{E_a}{RT}\right)$ { Increase A ⇒ better orient molecules for rxn; lower E_a ⇒ lower energy of Transition state



mostly lowers E_a

- catalysts can lead to multistep rxns → multiple peaks
- slowest step has highest peak in curve (hardest to get over)

Types of catalysts

Homogeneous: reactants, products & catalysts are in the same phase
ex. enzymes

Heterogeneous: reactants, products & catalysts are in diffrent phases
ex. solid catalyst for gas rxn

lecture 12

Equilibrium

Ex: Vapor Pressure or Acid-Base rxn

System reaches steady state

equilibrium if $rate_{fwd} = rate_{bwd}$

assume elementary rxn

$A \underset{k_{bwd}}{\overset{k_{fwd}}{\rightleftharpoons}} B$

$k_{fwd} [A]^{\alpha} \cdot [B]^{\beta} = k_{bwd} [C]^{\gamma} \cdot [D]^{\delta} \Leftrightarrow \frac{k_{fwd}}{k_{bwd}} = \frac{[C]^{\gamma}[D]^{\delta}}{[A]^{\alpha}[B]^{\beta}} = \frac{[Products]}{[Reactants]} = K_c$ = equilibrium constant, unitless!

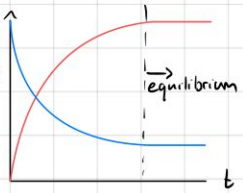
if elementary rxn: stochiometric coeff.

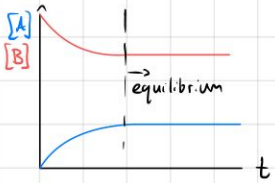
$K_c \equiv$ equilibrium constant for molarity concentrations

K_c [unitless]

is not constant $K_c(T)$!

use $[\frac{mol}{L}]$





equilibrium requires closed system (products & reactants can't escape)

Reactions in both directions still occurring, but at the same rate

so concentration doesn't change

! Does not Matter if we start with pure reactants or pure products

⇒ all rxns behave like elementary rxns near equilibrium → so unlike rate laws, we dont need to know rxn mechanism

$K_p = \frac{(P_C)^{\gamma}(P_D)^{\delta}}{(P_A)^{\alpha}(P_B)^{\beta}}$ ↑ unitless

partial Pressure: $P_i = \frac{n_i}{V} RT = [i]RT$ (concentration)

! unit [bar]

↳ to calculate use bar

$K_p = K_c (RT)^{\Delta n}$ | Δn = (#n of gaseous products) minus (#n of gaseous reactants)

$K_c \neq K_p$

K_c & K_p unitless because:

(Activity for substance in ideal mixture) → Ratio of Pressure or concentration to reference concentration or pressure

Activity of 1

$a_i = \frac{[i]}{1M}$

$a_i = \frac{P_i}{1 bar}$

$K_c = \frac{[Products]}{[reactants]}$ can ignore pure liquids & pure solids

if $K > 1$ Equil. to right mixture is mostly products

if $K < 1$ Equil. to left mixture is mostly reactants

rate = rate @ equil. but $[\text{reactants}]_{\text{concentration}} \neq [\text{products}]_{\text{concentration}}$!

Q = Momentaner Zustand

$$Q = \frac{[C]^{\gamma}[D]^{\delta}}{[A]^{\alpha}[B]^{\beta}}$$

to move towards equil. $Q = K_c$

$Q < K$: fwd rxn forms more products

$Q > K$: bwd rxn forms more reactants

reverse	$A \rightleftarrows B$	K_1	$B \rightleftarrows A$: $K_2 = \frac{1}{K_1}$
times n	$A \rightleftarrows B$	K_0	$3A \rightleftarrows 3B$: $K_2 = (K_0)^3$
multistep	$A \rightleftarrows B$ $B \rightleftarrows C$	K_1 K_2	$A \rightleftarrows C$: $K_3 = K_1 \cdot K_2$

K ≡ equil. Constant

homogeneous equil. all substances same phase

heterogeneous equil. substances different phase

Pure solids & pure liquids

In heterogeneous equilibria ⇒ exclude pure liquids & pure solids:

Ex: $PbCl_2(s) \rightleftarrows Pb^{2+}(aq) + 2Cl^-(aq)$

$K_c = [Pb^{2+}][Cl^-]^2$

Because concentration of pure (l) or (s) is constant, have activity of 1

For saturated solutions of low-solubility ionic solids: $K_c = K_{sp} \equiv$ **solubility Product** | Product of concentrations of dissolved Ions at equilibrium given in Brown book for many solids

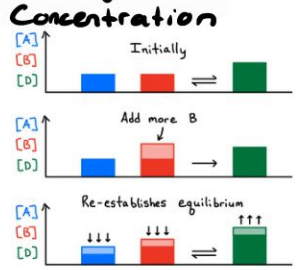
Le Châtelier-Principle

if sys. @ equil is disturbed by change in **T, P, concentration** equilibrium is shifted to counteract

can ignore pure solids & pure l

Change in Concentration

Ex: $A + B \rightleftharpoons D$ Initially at equilibrium



$$Q = \frac{[D]}{[A][B]} \overset{\text{concentrations}}{\overset{\text{cancel out}}{=}} \frac{n_D}{n_A n_B} = K_c$$ equilibrium

$$Q = \frac{n_D}{n_A (n_B + \Delta)} < K_c$$ { forward rxn forms more products

$$Q = \frac{n_D + x}{(n_A - x)(n_B + \Delta - x)} = K_c$$ equilibrium again

We wrote Q in terms of moles of each species. (Volumes cancel in concentration because all in same container)

See p. 9

Change in T

Treat heat (T) as product/reactant

Endothermic $A + T = B$ Exothermic $A = T + B$

T↗ : rxn moves right T↗ rxn moves left

T↘ : rxn moves left T↘ rxn moves right

because Ea is in $\exp(-\frac{E_a}{RT})$ → larger Ea & more strongly affected by change in Temp.

Change in Pressure

Ex: $N_2(g) + 3H_2(g) \rightleftarrows 2NH_3(g)$

System shifts equil. to counteract

$\Delta n = -2$ higher Pressure: → shifts right

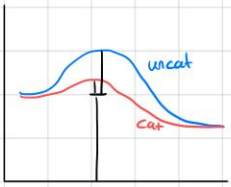
P↗ : equil. to side with less moles

P↘ : equil. to side with more moles

$K_p = \frac{(P_{NH_3})^2}{(P_{N_2})(P_{H_2})^3}$

$Q = \frac{2(P_{NH_3})^2}{(2P_{N_2})(2P_{H_2})^3} = \frac{1}{4}K_p$ $Q < K$

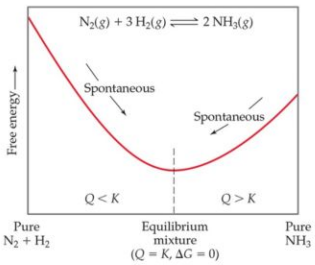
Catalyst effects on equilibria



catalysed rxn reach equilibrium faster

but K does not change due to cat.

$K_{fwd}^{cat} > K_{fwd}^{uncat}$, $K_{bwd}^{cat} > K_{bwd}^{uncat}$



$$\Delta G^\circ = -RT\ln(K) \Leftrightarrow K = \exp\left(\frac{-\Delta G}{RT}\right)$$

Gibbs

ΔG_f° (at STP, f = formation) = Free energy to form substance from elements

ΔG° = Free energy change for any rxn

$\Delta G < 0$, $K > 1$ equil. → products

$\Delta G > 0$, $K < 1$ equil. → reactants

$$\Delta G = \Delta G^\circ + RT\ln(Q)$$

at equilibrium

rxn spontaneous: $\Delta G < 0$

at equilibrium $\Delta G = 0$

T = absolute T

R = Gas const.

Q = rxn quotient

Q = K

Sample Problem

given: $[H_2]_i$, $[I_2]_i$, $[HI]_{equil.}$ search: K_c

$H_2 + I_2 \rightleftharpoons 2HI$

	H_2	I_2	$2HI$
initial concentration	$1 \cdot 10^{-3}$	$2 \cdot 10^{-3}$	0
change in concentration	$\overset{!}{-} 0,935 \cdot 10^{-3}$	$\overset{!}{-} 0,935 \cdot 10^{-3}$	$\overset{!}{+} 1,87 \cdot 10^{-3}$
equil. concentration	$0,065 \cdot 10^{-3}$	$1,065 \cdot 10^{-3}$	$1,87 \cdot 10^{-3}$

$$K_c = \frac{[HI]^2}{[H_2]^1 [I_2]^1}$$

1) change in $[CH]$: $1,87 \cdot 10^{-3} - 0 = 1,87^{-3}$
2) relate change to other rxn participants

$$\left(1,87 \cdot 10^{-3} \frac{\text{mol HI}}{\text{L}}\right)\left(\frac{1 \text{ mol } H_2}{2 \text{ mol HI}}\right) = 0,935 \cdot 10^{-3} \frac{\text{mol } H_2}{\text{L}}$$

$$\left(1,87 \cdot 10^{-3} \frac{\text{mol HI}}{\text{L}}\right)\left(\frac{1 \text{ mol } I_2}{2 \text{ mol HI}}\right) = 0,935 \cdot 10^{-3} \frac{\text{mol } I_2}{\text{L}}$$

3) fill out equil. concentration $[H_2] = 1 \cdot 10^{-3} - 0,935 \cdot 10^{-3} = 0,065 \cdot 10^{-3}$ mol/L

$[I_2] = 2 \cdot 10^{-3} - 0,935 \cdot 10^{-3} = 1,065 \cdot 10^{-3}$ mol/L

Sample Problem

given $[H_2]$, $[I_2]$, $K_c = 50,5$ (at given T)

$H_2 + I_2 \rightleftharpoons 2HI$

	H_2	I_2	$2HI$
initial concentration	1	2	0
change in concentration	$-x$	$-x$	$+2x$
equil. concentration	$1-x$	$2-x$	$0+2x$

1) Use stoichiometry to determine the range: for each mol of H_2 that reacts, one mol I_2 is consumed and two mol HI produced
2) use initial [c] and change in [c] by stoichiometry for equil. concentration
3) $K_c = \frac{[HI]^2}{[H_2][I_2]} = \frac{2x^2}{(1-x)(2-x)}$ → solve $x = 0,935$

Acid-Base Equilibria (S-B Reaktionen)

always SB pair, never alone

Acid ≡ gives H^+ in water increases $[H_3O^\oplus]$ concentration $HCl + H_2O \rightleftharpoons Cl^\ominus + H_3O^\oplus$

Base ≡ takes $H^\oplus$ in water increases $[OH^\ominus]$ concentration $NH_3 + H_2O \rightleftharpoons NH_4^\oplus + OH^\ominus$

$HCl(g) + NH_3(g) \rightleftharpoons NH_4Cl(s)$

Autoionisation of water

$$H_2O + H_2O \rightleftharpoons H_3O^\oplus + OH^\ominus \qquad K_c = \frac{[H_3O^+]^1 [OH^-]^1}{[\text{ignore pure liq } (=1)]} = K_w = 1,0 \cdot 10^{-14} \ (25°C) = [OH^-][H_3O^+]$$

when $[H_3O^+] = [OH^\ominus]$ solution is neutral

if $[H_3O^+]$ increases, $[OH^\ominus]$ decreases

$[H_3O^+] > [OH^-]$ "acidic"

$[H_3O] < [OH^-]$ "basic"

} always $[H_3O] \cdot [OH^-] = 1 \cdot 10^{-14}$

vice versa for bases

- Strong acid: gives up all H^+ to form H_3O^+, fully ionized
- weak acid: partially ionized, mix of HA & A^-
- HA without acidity: conjugate Base is stronger, accepts more H so ggw is left

generic acid "H+ acid"

$HA_{(aq)} + H_2O_{(l)} \rightleftharpoons A^-_{(aq)} + H_3O^+_{(aq)}$ $K_{acid} = \frac{[A^-][H_3O^+]}{[HA]}$ (H_2O left away bc pure)

Acid (gives H^+ away) — Base (takes H^+) — conjugate Base — conjugate Acid

generic Base

$B_{(aq)} + H_2O_{(l)} \rightleftharpoons HB^+_{(aq)} + OH^-_{(aq)}$ $K_{Base} = \frac{[HB^+][OH^-]}{[B]}$

Base (takes H^+) — Acid (gives H^+ away) — conjugate Acid — conjugate Base

GGW regel (Bircher)

search reactants on SB-Chart

draw immaginary line

- GGW right ⇌
- GGW left ⇌

each Base has a conjugate acid

Acid has a conjugate base

the stronger the acid, the weaker the conjugate base GS

pH - Scale

$p\{\} = -\log\{\}$

$$pH = -\log_{10}[H_3O^+] \quad <=> \quad [H_3O^+][OH^-] = 1\cdot10^{-14}$$

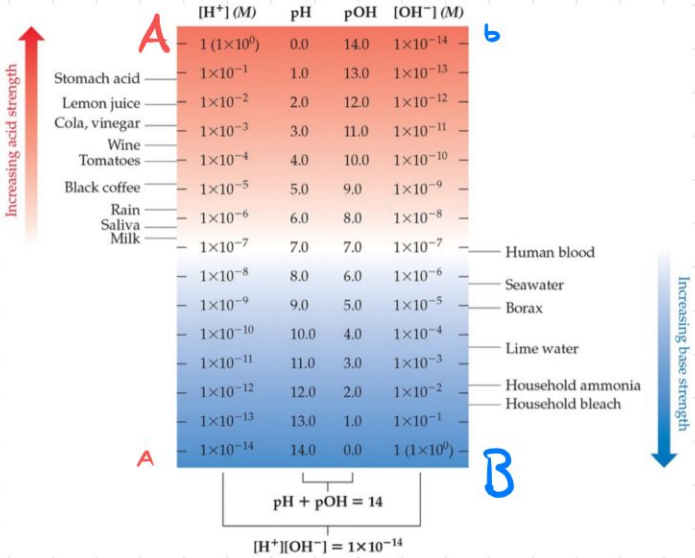
$$pKw = -\log\{Kw\} = 14 = -\log[OH^-][H_3O^+] = 14$$

pH 7 = neutral
pH > 7 Basic
pH < 7 Acidic

if $[H_3O^+] = 1\cdot10^{-3}$
then $[OH] = 1\cdot10^{-11}$
$(-3 - 11 = -14)$

$$pOH = -\log[OH^-], \quad pH = -\log[H_3O^+]$$

$$pH + pOH = 14$$



$$K_a = \frac{[A^-][H_3O^+]}{[HA]} \qquad pK_a = -\log(K_a)$$

The larger K_a, the stronger acid
The smaller pK_a → the stronger acid

$$K_b = \frac{[HB^+][OH^-]}{[B]} \qquad pK_b = -\log(K_b)$$

The larger K_b, the stronger base
The smaller pK_b → the stronger base

When $pH = pK_a$

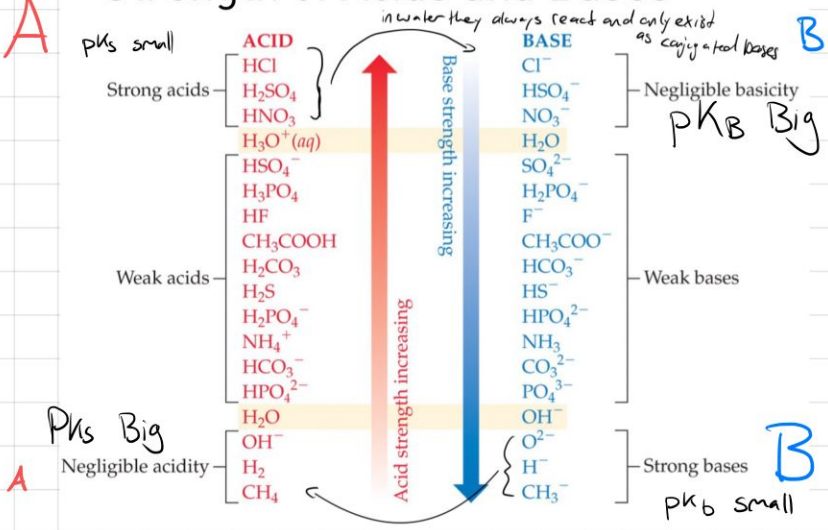
$$-\log[H_3O^+] = -\log\left[\frac{[A^-][H_3O^+]}{[HA]}\right] \Rightarrow [HA] = [A^-]$$

$pH = pK_s$ Acid half ionized

$pH > pK_s$ Acid mostly exists as A^-

$pH < pK_s$ Acid mostly exists as HA

Strength of Acids and Bases



Explains (Brown, p. 762):

"The ions H_3O^+ and OH^- are, respectively, the strongest possible acid and strongest possible base that can exist at equilibrium in aqueous solution."

"Stronger acids react with water to produce H_3O^+ ions, and stronger bases react with water to produce OH^- ions, a phenomenon known as the leveling effect."

These H_3O^+ and OH^- ions are stronger acids and bases than most other chemicals. Thus, can give/take H^+.

⇒ strong acids are immediately converted into H_3O^+ and their conjugate pair respectively. so the strongest acid in an aqueous soln can ever only be H_3O^+. Same goes for bases and OH^-

Common-Ion effect

$$CH_3COOH(aq) + H_2O(l) \rightleftharpoons CH_3COO^-(aq) + H_3O^+(aq)$$

add strong electrolyte: CH_3COONa

$$CH_3COONa(aq) + H_2O(l) \rightleftharpoons CH_3COO^-(aq) + Na^+(aq)$$

⇒ shifts equilibrium left, decreasing $[H_3O^+]$
⇒ equilibrium constant does not change, only relative concentration of products & reactants

Common Ion effect

Whenever a weak electrolyte and a strong electrolyte containing a common Ion, are together in a soln, the weak Electrolyte ionizes less, than if it were alone in a solution

Buffers

Protect against pH disturbances and maintain pH (over a range)

⇒ weak conjugate acid/base pair protects against $[H_3O^+]$ or $[OH^-]$

$$CH_3COOH + H_2O \rightleftharpoons CH_3COO^- + H_3O^+$$

$$CH_3COO^- + H_3O^+ \rightleftharpoons CH_3COOH + H_2O \quad \text{Disturbance extra } H_3O^+$$

$$CHCOOH + OH^- \rightleftharpoons CH_3COO^- + H_2O \quad \text{Disturbance extra } OH^-$$

$$[H_3O^+] = K_s \frac{[HA]}{[A^-]} \Big\} \text{sets pH}$$

as long as Disturbance is small relative to $[HA][A^-]$
⇒ small pH change

calculate Buffer

$$pH = pK_a + \log \frac{[Base]}{[acid]}$$

can tune Buffer with common-ion effect (add MA) to increase $[A^-]$
To effectively resist pH changes in both directions
$[HA] = [A^-] \Leftrightarrow pH = pK_a$
Blood is buffer too, using H_2CO_3, HCO_3^-

Brønsted-Lowry acid-base theory

A bronsted lowry acid is any species that can donate H^+ to another molecule. Bronsted base is the H^+ acceptor. Doesnt need to be water

Sample Problem: given: $[HCOOH] = 0{,}1\ M$, $pH = 2{,}38$, aq. soln calculate K_a.

$HCOOH + H_2O \rightleftharpoons HCOO^- + H_3O^+$, $pH = 2{,}38$

$$K_a = \frac{[H_3O^+][HCOO^-]}{[HCOOH]} \qquad pH = -\log_{10}[H_3O^+] = 2{,}38 \Leftrightarrow [H_3O^+] = 10^{-2{,}38} = 4{,}2 \cdot 10^{-3}\ M$$

ignore autoionisation of water

	HCOOH	⇌	H_3O^+	$HCOO^-$
[c] init	0,1		0	0
delta [c]	$-4{,}2 \cdot 10^{-3}$		$+4{,}2 \cdot 10^{-3}$	$+4{,}2 \cdot 10^{-3}$
equil. [c]	$0{,}1 - 4{,}2 \cdot 10^{-3}$		$4{,}2 \cdot 10^{-3}$	$4{,}2 \cdot 10^{-3}$

initial concentration was 0, so this must be the change in concentration of H_3O^+

$$\Rightarrow K_a \frac{[H_3O^+][HCOO^-]}{[HCOOH]} = \frac{(4{,}2 \cdot 10^{-3}) \cdot (4{,}2 \cdot 10^{-3})}{0{,}1} = 1{,}8 \cdot 10^{-4}$$

Sample 2 given: $[HCN] = 0{,}2\ M$, calculate pH

$$HCN + H_2O \rightleftharpoons CN^- + H_3O^+ \qquad K_a = \frac{[CN^-][H_3O^+]}{[HCN]} = 4{,}9 \cdot 10^{-10}$$

	HCN	⇌	H_3O^+	CN^-
initial [c]	0,2		0	0
delta [c]	$-x$		$+x$	$+x$
equil. [c]	$(0{,}2 - x)$		x	x

⇓ substitute concentrations with x

$$K_a = \frac{x \cdot x}{0{,}2 - x} = 4{,}9 \cdot 10^{-10}$$

⇒ solve for x: $x = 9{,}9 \cdot 10^{-6}\ M = [H_3O^+]$

⇒ $pH = -\log[H_3O^+] = -\log(9{,}9 \cdot 10^{-6}) = \underline{5{,}00}$

Electrochemistry (Redox)

LEO the Lion says GER

Ox Numbers

Atom with higher EN takes e^- ⇒ sum of Ox Numbers equals net charge

Atoms in elemental form: 0

Monoatomic Ions: ionic charge

Nonmetals in ionic/molecular compounds negative Ox Number

O	-II	exept O_2^{2-} -I
H	+I	exept bonded to metal -I
F	-I	always
Cl, Br, I	-I	exept bonded to hydrogen



Ox Number
↘ Reduction, takes e^-
↗ Oxidation, gives e^-

$$\overset{0}{Mg}(s) + 2H\overset{+I\ -I}{Cl}(aq) \rightarrow \overset{+II\ -I}{MgCl}(aq) + \overset{0}{H_2}(aq)$$

Mg oxidized Mg reduces H; Mg is reductant

H reduced : H oxidizes Mg ; H is oxidizer

Procedure for aqueous soln

1) Divide rxn into oxidation half reaction and reduction half reaction
2) Balance Elements other than H and O
   Balance O atoms by adding H_2O as needed
   Balance H atoms by adding H^+ as needed
   Balance charge by adding e^- as needed
3) Multiply two half rxn by Integers to equate e^- lost & Gained ← by adding H_2O
4) Add half rxn and if possible cancel species appearing on both sides ← by adding H^+
5) Check to make sure atoms and charges are balanced.

permanganate ion + oxalate ion

Note: this reaction is not balanced yet. We just wrote down "skeleton equation"

Example Balance $MnO_4^- + C_2O_4^{2-} \rightarrow Mn^{2+} + CO_2$ in acidic H_2O

	Reduction	Oxidation
Half-rxns:	$MnO_4^- \rightarrow Mn^{2+}$ (+7 -2 → +2)	$C_2O_4^{2-} \rightarrow CO_2$ (+3 -2 → +4 -2)
Balance non H,O	$MnO_4^- \rightarrow Mn^{2+}$	$C_2O_4^{2-} \rightarrow 2CO_2$
Balance O	$MnO_4^- \rightarrow Mn^{2+} + 4H_2O$	$C_2O_4^{2-} \rightarrow 2CO_2$
Balance H	$MnO_4^- + 8H^+ \rightarrow Mn^{2+} + 4H_2O$	$C_2O_4^{2-} \rightarrow 2CO_2$
By adding e^-'s: Balance charge	$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$	$C_2O_4^{2-} \rightarrow 2CO_2 + 2e^-$
Multiply half-rxns by integers to balance electrons lost and gained in reduction and oxidation half-rxns: Balance e^-'s lost gained	$2MnO_4^- + 16H^+ + 10e^- \rightarrow 2Mn^{2+} + 8H_2O$	$5C_2O_4^{2-} \rightarrow 10CO_2 + 10e^-$
Add half-rxns, Canceling species appearing on both sides of equation: Combine half-rxns	$2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$	

Balanced!

Are atoms and charges balanced? Check balance

Atoms: 2 Mn ✓, 10 C ✓, 20 oxygen ✓, 16 H ✓

Charge: left-hand side $-2-10+16=+4$ ✓ right-hand side $+4$

What if soln is basic?

1) Balance half rxns as if they occur in acidic soln
2) Count Number of H^+ in each half rxn, and add same number of OH^- to each side
   → neutralize H^+ with OH^-
   → equal H^+ and OH^- on same side form H_2O
   → cancel out any H_2O that is on both sides
3) Multiply two half rxn by Integers to equate e^- lost & Gained ← by adding H_2O
4) Add half rxn and if possible cancel species appearing on both sides ← by adding H^+
5) Check to make sure atoms and charges are balanced.

Again: this reaction is not balanced yet. We just wrote down "skeleton equation"

cyanate ion

Example Balance $CN^- + MnO_4^- \rightarrow CNO^- + MnO_2(s)$ in basic H_2O

Proceed as if reaction is in acidic solution

	Reduction	Oxidation
Half-rxns:	$MnO_4^- \rightarrow MnO_2$ (+7 -2 → +4 -2)	$CN^- \rightarrow CNO^-$ (+2 -3 → +4 -3 -2)
Balance non H,O	$MnO_4^- \rightarrow MnO_2$	$CN^- \rightarrow CNO^-$
Balance O	$MnO_4^- \rightarrow MnO_2 + 2H_2O$	$CN^- + H_2O \rightarrow CNO^-$
Balance H	$MnO_4^- + 4H^+ \rightarrow MnO_2 + 2H_2O$	$CN^- + H_2O \rightarrow CNO^- + 2H^+$
By adding e^-'s: Balance charge	$MnO_4^- + 4H^+ + 3e^- \rightarrow MnO_2 + 2H_2O$	$CN^- + H_2O \rightarrow CNO^- + 2H^+ + 2e^-$
Add OH^- to both sides — enough to neutralize H^+	$MnO_4^- + 4H^+ + 3e^- + 4OH^- \rightarrow MnO_2 + 2H_2O + 4OH^-$	$CN^- + H_2O + 2OH^- \rightarrow CNO^- + 2H^+ + 2e^- + 2OH^-$
$H^+ + OH^- \rightarrow H_2O$: Neutralize H^+	$MnO_4^- + 4H_2O + 3e^- \rightarrow MnO_2 + 2H_2O + 4OH^-$	$CN^- + H_2O + 2OH^- \rightarrow CNO^- + 2H_2O + 2e^-$
Cancel H_2O on both sides: Cancel H_2O	$MnO_4^- + 2H_2O + 3e^- \rightarrow MnO_2 + 4OH^-$	$CN^- + 2OH^- \rightarrow CNO^- + H_2O + 2e^-$

Then proceed as in acidic case

Multiply half-rxns by integers to balance electrons lost and gained in reduction and oxidation half-rxns: Balance e^-'s lost gained

$2MnO_4^- + 4H_2O + 6e^- \rightarrow 2MnO_2 + 8OH^-$ · 2 (Multiply by 2)

$3CN^- + 6OH^- \rightarrow 3CNO^- + 3H_2O + 6e^-$ (Multiply by 3)

Add half-rxns, Canceling species appearing on both sides of equation: Combine half-rxns

$2MnO_4^- + H_2O + 3CN^- \rightarrow 2MnO_2 + 3CNO^- + 2OH^-$

Balanced!

Are atoms and charges balanced? Check balance

Atoms: 2 Mn ✓, 3 C ✓, 3 N ✓, 9 oxygen ✓, 2 H ✓

Charge: left-hand side $-2-3=-5$ ✓ right-hand side $-3-2=-5$

Voltaic Cells

Redox is spontaneous if $\Delta G < 0$

Spontaneous if the other metal is below it

$Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$

Redox Regel (Bircher)

search reactants on Redox table

draw immaginary line

↘ rxn is spontaneous

↙ rxn is not spontaneous

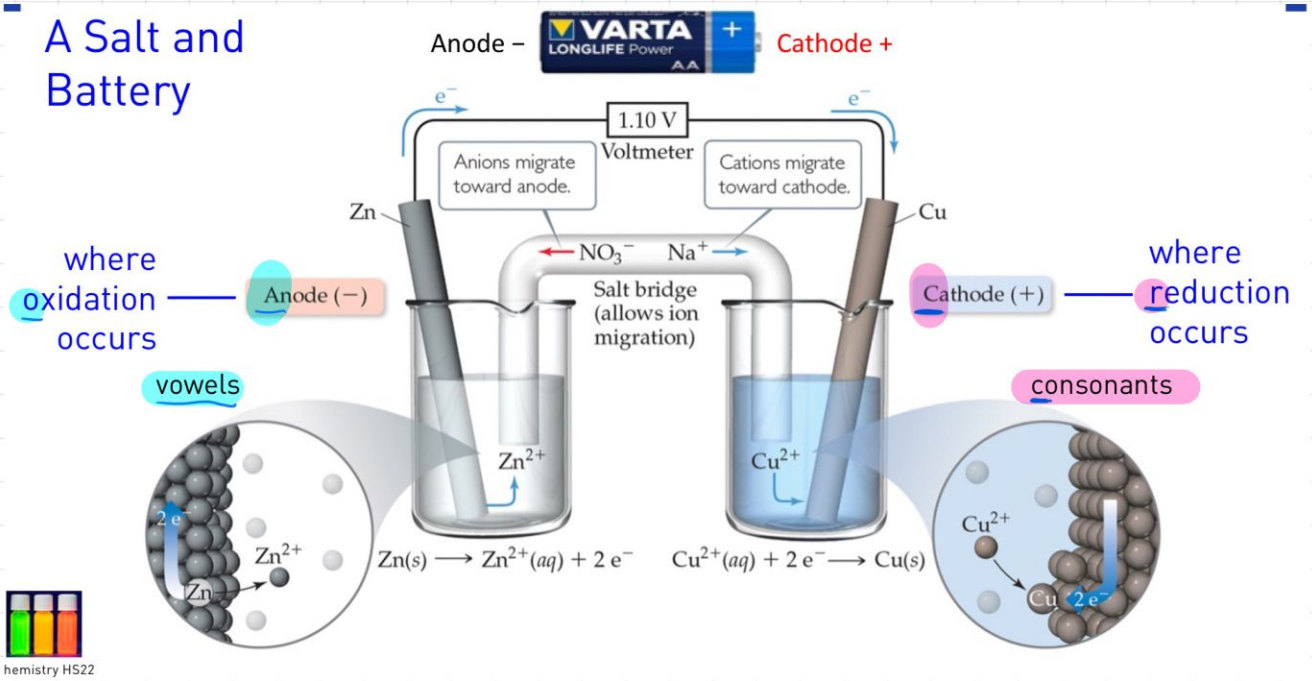
TABLE 4.5 Activity Series of Metals in Aqueous Solution

Metal	Oxidation Reaction
Lithium	$Li(s) \longrightarrow Li^+(aq) + e^-$
Potassium	$K(s) \longrightarrow K^+(aq) + e^-$
Barium	$Ba(s) \longrightarrow Ba^{2+}(aq) + 2e^-$
Calcium	$Ca(s) \longrightarrow Ca^{2+}(aq) + 2e^-$
Sodium	$Na(s) \longrightarrow Na^+(aq) + e^-$
Magnesium	$Mg(s) \longrightarrow Mg^{2+}(aq) + 2e^-$
Aluminum	$Al(s) \longrightarrow Al^{3+}(aq) + 3e^-$
Manganese	$Mn(s) \longrightarrow Mn^{2+}(aq) + 2e^-$
Zinc	$Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^-$
Chromium	$Cr(s) \longrightarrow Cr^{3+}(aq) + 3e^-$
Iron	$Fe(s) \longrightarrow Fe^{2+}(aq) + 2e^-$
Cobalt	$Co(s) \longrightarrow Co^{2+}(aq) + 2e^-$
Nickel	$Ni(s) \longrightarrow Ni^{2+}(aq) + 2e^-$
Tin	$Sn(s) \longrightarrow Sn^{2+}(aq) + 2e^-$
Lead	$Pb(s) \longrightarrow Pb^{2+}(aq) + 2e^-$
Hydrogen	$H_2(g) \longrightarrow 2H^+(aq) + 2e^-$
Copper	$Cu(s) \longrightarrow Cu^{2+}(aq) + 2e^-$
Silver	$Ag(s) \longrightarrow Ag^+(aq) + e^-$
Mercury	$Hg(l) \longrightarrow Hg^{2+}(aq) + 2e^-$
Platinum	$Pt(s) \longrightarrow Pt^{2+}(aq) + 2e^-$
Gold	$Au(s) \longrightarrow Au^{3+}(aq) + 3e^-$

Ease of oxidation increases

any metal can be oxidized by element below it

A Salt and Battery



Electrodes

Solid Metals connected to external circuit

In Cu-Zn example, electrode materials participate in rxn but not always.

Zn Anode losing mass (gets dissolved) Cu Cathode gaining mass

EMF = Maximal Potential Difference between two electrodes

Electrolyte

Liquid with ions that can conduct electric current

To maintain charge neutrality, ions need to move in electrolyte

Salt bridge

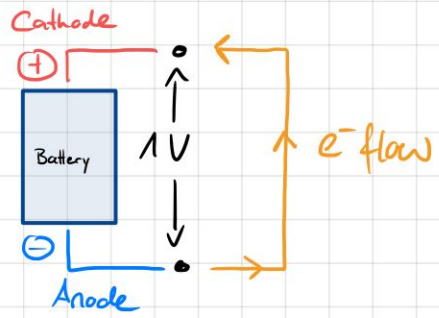
Tube containing electrolyte solution, allowing ion migration

Electrolyte can be a gel to stay in tube

Anions move toward Anode, cations move towards cathode

Energy and Batteries

$$1\,V = \frac{1{,}6\cdot10^{-19}\,J}{1{,}6\cdot10^{-19}\,J}\left[\frac{\text{energy}}{\text{charge}}\right] \qquad 1\,V = 1\,\frac{J}{C}$$



Voltage is a potential Energy difference per charge when we compare two points in a Circuit. Here we are comparing cathode to anode on a one volt battery

Cell Potentials

Cell Potential:

$$E_{cell} = E^\circ_{red}(\text{cathode}) - E^\circ_{red}(\text{anode})$$

reduction is reverse of oxidation. minus because oxidation actually occurs at anode

E°_{cell} = Voltage of cell at STP

- to be useful $E^\circ_{cell} > 0$ (by convention)
- useful means e^- flow spontaneously from Anode to Cathode to do work
- Electrons have higher Potential at Anode and flow to Cathode
- E°_{cell} is "intensive" quantity - does not depend on size of system

E°_{red} = Potential Energy available if reduced

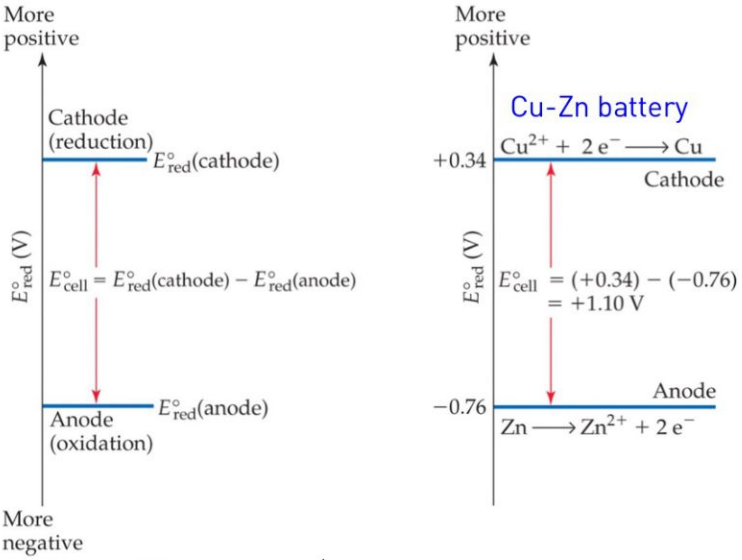
- Half cell potential cannot be measured directly, because we need full cell to measure something
- But can Determine E°_{red} for any half reaction by measuring with SHE (standard Hydrogen Electrode)
- All half-cell potentials Listed as reducing potentials, but half-rxns are reversible, depending on what half-cell it is connected to. More positive E°_{red} means greater tendency for reduction

TABLE 20.1 Standard Reduction Potentials in Water at 25°C

E°_{red} (V)	Reduction Half-Reaction
+2.87	$F_2(g) + 2\,e^- \longrightarrow 2\,F^-(aq)$
+1.51	$MnO_4^-(aq) + 8\,H^+(aq) + 5\,e^- \longrightarrow Mn^{2+}(aq) + 4\,H_2O(l)$
+1.36	$Cl_2(g) + 2\,e^- \longrightarrow 2\,Cl^-(aq)$
+1.33	$Cr_2O_7^{2-}(aq) + 14\,H^+(aq) + 6\,e^- \longrightarrow 2\,Cr^{3+}(aq) + 7\,H_2O(l)$
+1.23	$O_2(g) + 4\,H^+(aq) + 4\,e^- \longrightarrow 2\,H_2O(l)$
+1.06	$Br_2(l) + 2\,e^- \longrightarrow 2\,Br^-(aq)$
+0.96	$NO_3^-(aq) + 4\,H^+(aq) + 3\,e^- \longrightarrow NO(g) + 2\,H_2O(l)$
+0.80	$Ag^+(aq) + e^- \longrightarrow Ag(s)$
+0.77	$Fe^{3+}(aq) + e^- \longrightarrow Fe^{2+}(aq)$
+0.68	$O_2(g) + 2\,H^+(aq) + 2\,e^- \longrightarrow H_2O_2(aq)$
+0.59	$MnO_4^-(aq) + 2\,H_2O(l) + 3\,e^- \longrightarrow MnO_2(s) + 4\,OH^-(aq)$
+0.54	$I_2(s) + 2\,e^- \longrightarrow 2\,I^-(aq)$
+0.40	$O_2(g) + 2\,H_2O(l) + 4\,e^- \longrightarrow 4\,OH^-(aq)$
+0.34	$Cu^{2+}(aq) + 2\,e^- \longrightarrow Cu(s)$
0 [defined] (SHE)	$2\,H^+(aq) + 2\,e^- \longrightarrow H_2(g)$
−0.28	$Ni^{2+}(aq) + 2\,e^- \longrightarrow Ni(s)$
−0.44	$Fe^{2+}(aq) + 2\,e^- \longrightarrow Fe(s)$
−0.76	$Zn^{2+}(aq) + 2\,e^- \longrightarrow Zn(s)$
−0.83	$2\,H_2O(l) + 2\,e^- \longrightarrow H_2(g) + 2\,OH^-(aq)$
−1.66	$Al^{3+}(aq) + 3\,e^- \longrightarrow Al(s)$
−2.71	$Na^+(aq) + e^- \longrightarrow Na(s)$
−3.05	$Li^+(aq) + e^- \longrightarrow Li(s)$

ETH zürich

Determining Cell Potentials



From Table 20.1

+0.34 $Cu^{2+}(aq) + 2\,e^- \longrightarrow Cu(s)$

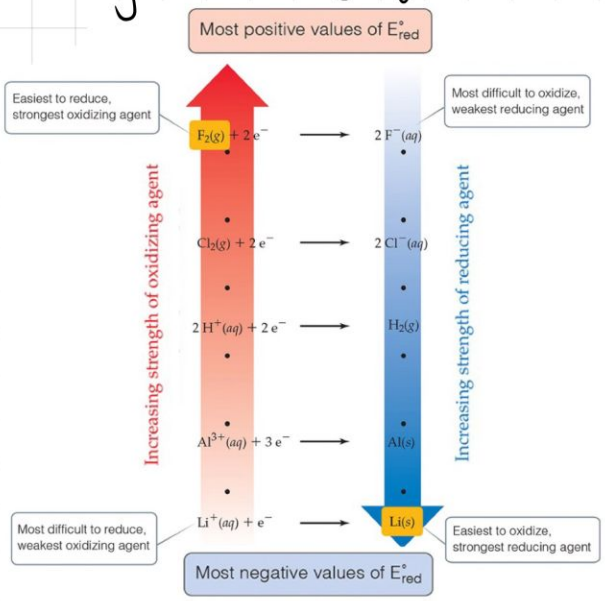


−0.76 $Zn^{2+}(aq) + 2\,e^- \longrightarrow Zn(s)$

Chemistry HS22

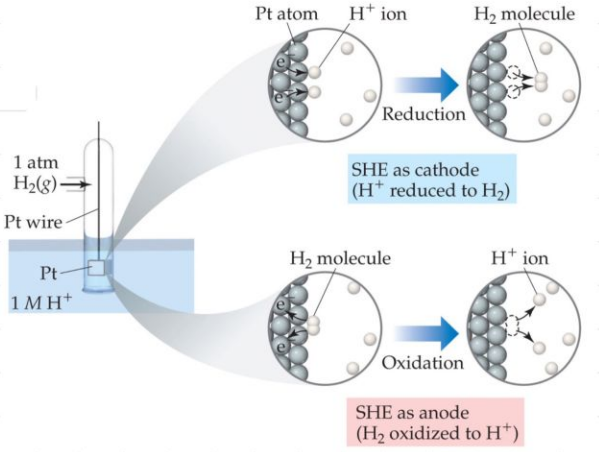
Voltaic cell = Galvanic cell

Oxidizing and reducing agents



Standard Hydrogen Electrode

SHE · at STP
· can operate as anode or cathode
· Depends on other half-cell



Gibbs

$$\Delta G^\circ = -nFE^\circ_{cell}$$

$\Delta G^\circ = \frac{J}{mol}$ of rxn

F = Faraday's constant
= 96,485 $\frac{C}{mol}$ of e^-

n = positive unitless number
moles of e^- transferred in balanced cell rxn

if $E^\circ_{cell} > 0$, $\Delta G^\circ < 0$

Spontaneous

check units $\frac{J}{mol} = -\frac{C}{mol} \cdot V = \frac{J}{mol}$

$C = \frac{J}{V}$

Big Picture

- By combining half-cell rxns, we can design battery
- Lithium is mostly oxidized; $E^\circ_{red} = -3,05\,V$
  ↳ makes sense → used in normal batteries
- if instead of extracting work, we apply external potential (Voltage)
- recharge batteries by "reversing" chemistry

Related topics: fuel cells, electroplating, corrosion